

NIVESH BANK

Invest. Grow. Prosper.

Enterprise Microservices Architecture

Spring Boot 3.x | Java 21 | Cloud Native | Event-Driven

14
Microservices

200+
REST Endpoints

21
Kafka Topics

50K
Peak TPS

99.99%
Uptime SLA

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Principal Architect | 25 Years Experience

CONFIDENTIAL | INTERNAL ARCHITECTURE DOCUMENT | NOT FOR DISTRIBUTION

Java 21

Spring Boot 3.x

Kubernetes

Apache Kafka

PostgreSQL

Redis

Cassandra

MongoDB

Elasticsearch

AWS EKS

Executive Summary

This document presents the complete microservices architecture design for Nivesh Bank’s next-generation core banking platform. The system is architected to serve 500+ million customers, process 10 million+ daily transactions, and maintain 99.99% uptime SLA.

Architecture Philosophy: The platform follows Domain-Driven Design (DDD) where each bounded context maps to an independent microservice. Services communicate asynchronously via Apache Kafka for high-throughput operations and synchronously via REST/gRPC for real-time queries. The entire platform runs on Kubernetes (AWS EKS) with auto-scaling, blue-green deployments, and zero-downtime upgrades.

Metric	Target	Technology
Daily Active Users	50 Million	API Gateway + CDN
Peak TPS	50,000 TPS	Kafka + In-Memory Cache
Account Records	500+ Million	Sharded PostgreSQL + Cassandra
Availability SLA	99.99% (52 min downtime/year)	Multi-AZ Kubernetes
RTO	< 30 seconds	Standby replicas + Chaos Engineering
RPO	< 5 seconds	Synchronous DB replication
API Latency (p95)	< 200ms	Redis caching + Connection pooling
Data Encryption	AES-256 + TLS 1.3	AWS KMS + Vault

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2 Architecture Diagram

The following diagram illustrates the five-layer architecture of Nivesh Bank's platform, from client-facing interfaces through the API gateway, service mesh, data layer, and infrastructure. All 14 microservices are shown with their port assignments and the shared Apache Kafka event bus that connects them asynchronously.

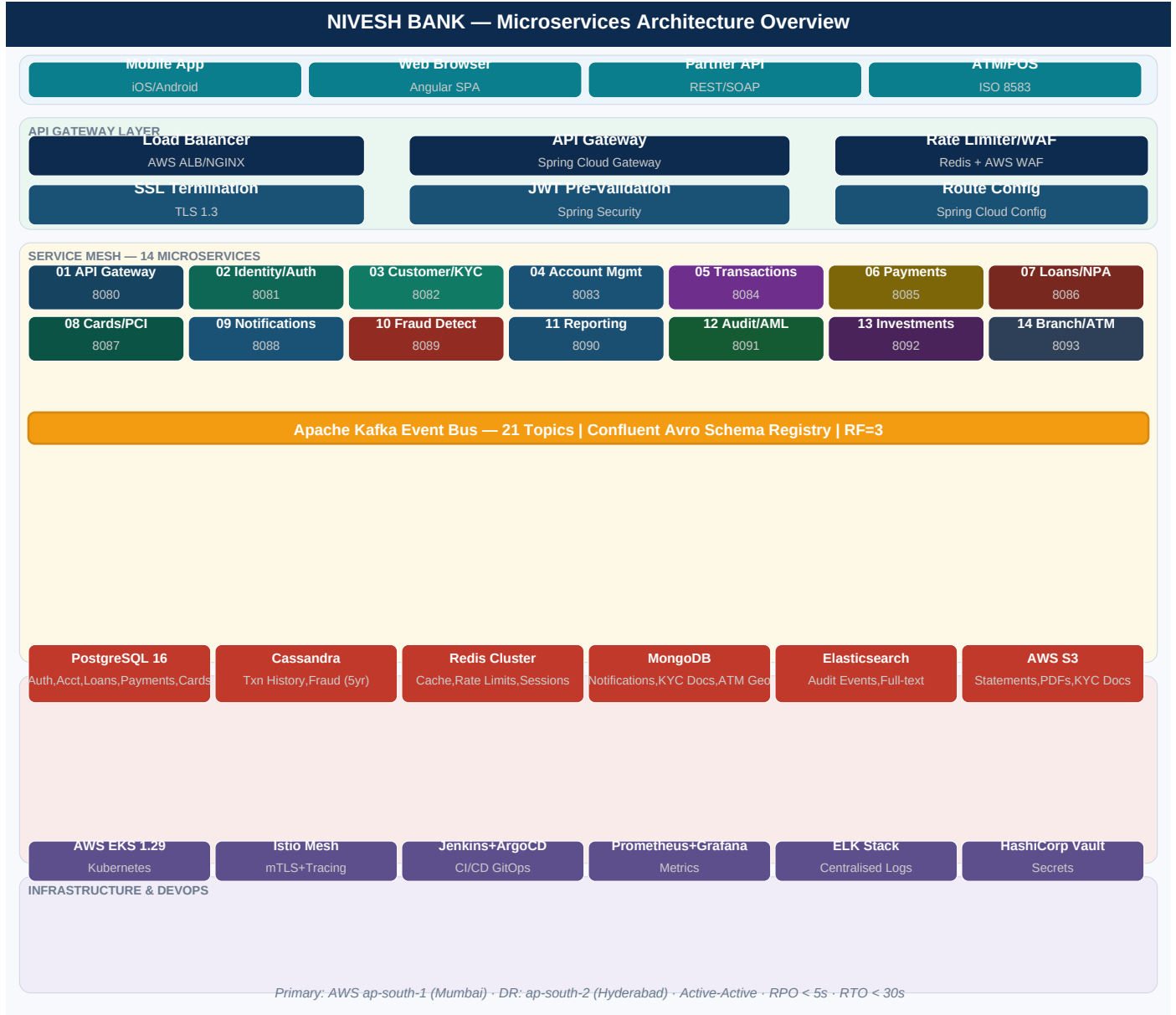


Figure 1: Nivesh Bank Enterprise Microservices Architecture — Full System View

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System Architecture Overview

Architectural Layers

Layer	Components	Technology Stack
Client Layer	Mobile App (iOS/Android), Web Browser, Partner APIs, ATM/POS Terminals	React Native, Angular, REST/SOAP
API Gateway Layer	Load Balancer, API Gateway, Rate Limiter, SSL Termination	Spring Cloud Gateway, NGINX, AWS ALB
Service Mesh Layer	14 Microservices, Service Discovery, Circuit Breaker	Kubernetes, Istio, Resilience4j
Data Layer	Relational DBs, NoSQL, Cache, Message Broker, Object Store	PostgreSQL, Cassandra, Redis, Kafka, S3
Infrastructure Layer	Container Orchestration, CI/CD, Monitoring, Secret Management	AWS EKS, Jenkins, Grafana, HashiCorp Vault

Microservice Catalogue

#	Service Name	Domain	Primary DB	Port
01	API Gateway Service	Infrastructure	Redis	8080
02	Identity & Authentication Service	Security	PostgreSQL	8081
03	Customer Profile Service	CRM	PostgreSQL + MongoDB	8082
04	Account Management Service	Core Banking	PostgreSQL	8083
05	Transaction Processing Service	Core Banking	PostgreSQL + Cassandra	8084
06	Payments & Fund Transfer Service	Payments	PostgreSQL	8085
07	Loan & Credit Management Service	Lending	PostgreSQL	8086
08	Card Management Service	Cards	PostgreSQL	8087
09	Notification Service	Communication	MongoDB	8088
10	Fraud Detection Service	Risk & Compliance	Redis + Cassandra	8089
11	Reporting & Statement Service	Analytics	PostgreSQL + S3	8090
12	Audit & Compliance Service	Compliance	PostgreSQL + Elasticsearch	8091
13	Investment & Wealth Service	Wealth Management	PostgreSQL	8092
14	Branch & ATM Management Service	Operations	PostgreSQL + MongoDB	8093

4 Infrastructure & Cross-Cutting Concerns

Spring Boot 3.x Common Dependencies (All Services)

- **spring-boot-starter-web** — Embedded Tomcat, REST endpoints
- **spring-boot-starter-data-jpa** — Hibernate ORM, repository pattern
- **spring-boot-starter-security** — Security filter chain base
- **spring-boot-starter-oauth2-resource-server** — JWT validation
- **spring-boot-starter-actuator** — Health, metrics, info endpoints
- **spring-cloud-starter-netflix-eureka-client** — Service discovery registration
- **spring-cloud-starter-openfeign** — Declarative REST client between services
- **spring-kafka** — Kafka producer/consumer integration
- **resilience4j-spring-boot3** — Circuit breaker, retry, rate limiter
- **micrometer-registry-prometheus** — Metrics export to Prometheus/Grafana
- **springdoc-openapi-starter-webmvc-ui** — Auto-generated Swagger/OpenAPI docs
- **lombok** — Boilerplate reduction (@Data, @Builder, @Slf4j)
- **mapstruct** — Entity ↔ DTO mapping
- **flyway-core** — Database schema versioning and migration

Cross-Cutting Patterns Applied

Pattern	Implementation	Why
Saga Pattern	Choreography via Kafka events	Distributed transactions without 2PC
CQRS	Separate Read/Write models per service	Optimise read-heavy banking queries independently
Circuit Breaker	Resilience4j @CircuitBreaker	Prevent cascade failures in service mesh
Outbox Pattern	Transactional outbox + Debezium CDC	Guarantee event delivery without dual-write
Idempotency	Idempotency-Key header + DB unique constraint	Safe retry of payment/transfer APIs
API Versioning	URI versioning /api/v1/ /api/v2/	Backward compatibility during evolution
Distributed Tracing	Micrometer Tracing + Zipkin/Jaeger	End-to-end request correlation
Secrets Management	HashiCorp Vault + AWS KMS	Zero hardcoded credentials, dynamic rotation
Config Management	Spring Cloud Config Server + Git	Centralised, environment-specific configs

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IFSC Code Generation Logic (Nivesh Bank)

Every Indian bank branch has an 11-character IFSC (Indian Financial System Code) used by NEFT, RTGS, and IMPS for routing. Nivesh Bank's IFSC structure is deterministic, collision-free, and RBI-compliant.

IFSC Structure: N V S H 0 X X X X X X

Entity: IFSC Code Breakdown

Each component of the 11-character IFSC code and its derivation rule.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
Chars 1–4	Bank Code fixed identifier for Nivesh Bank, assigned by RBI. Cannot change post-registration.	NVSH	Always "NVSH". Hardcoded constant.
Char 5	Reserved — RBI mandates this character always be literal "0" (zero) for all Indian banks.	0	Always literal "0". Regulatory requirement.
Chars 6–8	Region Code — first 3 digits of the branch's PIN code (postal zone). Uniquely identifies the postal circle / metro zone.	400 (for PIN 400001)	Derived from branch PIN at registration. Read-only.
Chars 9–11	Branch Sequence — 3-digit zero-padded sequential counter within the region. Max 999 branches per PIN zone. Only 1 branch per 5km radius is registered; uniqueness enforced via PostGIS ST_DWithin check at onboarding.	001	DB sequence per region. Never recycled. Guarantees uniqueness.

Example IFSC derivations:

- Mumbai (PIN 400001) → NVSH0400001 (first Mumbai branch)
- Bangalore (PIN 560034) → NVSH0560001 (first Bangalore branch)
- Pune (PIN 411001) → NVSH0411001
- Chennai (PIN 600020) → NVSH0600001
- Hyderabad (PIN 500032) → NVSH0500001

Collision prevention: A unique constraint on (region_code, branch_sequence) in the Branch table prevents duplicate IFSC. BranchOnboardingService validates: (a) no existing branch within 5km radius (PostGIS ST_DWithin check), (b) sequence auto-incremented from DB sequence per region_code.

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Account Number Generation (11-Digit Numeric)

Account numbers at Nivesh Bank are **11-digit numeric strings**. They are human-readable, Luhn-checksum validated, and embed enough structure for routing and fraud detection.

Entity: Account Number Structure (11 Digits)

All Nivesh Bank account numbers are 11-digit numerics encoding type, region, sequence, and a Luhn check digit.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
Digit 1	Account Type Prefix — encodes the product type for instant visual identification. Non-zero.	1=SAVINGS, 2=CURRENT, 3=FD, 4=RD, 5=LOAN, 6=NRI, 7=SALARY, 8=JAN_DHAN	ENUM via AccountTypePrefix. Always non-zero.
Digits 2–4	Region Code — first 3 digits of the branch IFSC region segment (matches PIN zone).	400 (Mumbai), 560 (Bangalore), 411 (Pune)	Derived from branch IFSC at account opening. Immutable.
Digits 5–10	Unique Sequential Counter — 6-digit zero-padded sequence from a DB global sequence per (type_prefix + region_code). Never reused even after account closure.	000001 to 999999	DB sequence per partition. Never recycled.
Digit 11	Luhn Check Digit — computed using Luhn algorithm over digits 1–10. Enables offline validation before any network call.	7 (computed)	Computed server-side. Invalid if check fails on inbound APIs.

Examples:

- 14000000017 — Savings (1), Mumbai region (400), sequence 000001, Luhn digit 7
- 25600000013 — Current (2), Bangalore region (560), sequence 000001, Luhn digit 3

- **34110000425** — Fixed Deposit (3), Pune region (411), sequence 000042, Luhn digit 5
- **55000012349** — Loan (5), Hyderabad region (500), sequence 001234, Luhn digit 9

01 API Gateway Service

Single entry point · Rate limiting · Routing · JWT pre-validation

Purpose & Responsibility

The API Gateway is the single entry point for ALL external traffic — mobile apps, web browsers, ATM networks, partner bank APIs, and government systems (NPCI, NACH, GST). It performs SSL termination, JWT token pre-validation, route-based load balancing, rate limiting, request/response transformation, and API versioning. No downstream service is ever directly exposed to the internet.

Architectural Rationale

Banking systems require a unified security perimeter. Without a gateway, each service would independently handle SSL, rate limiting, and CORS — creating inconsistency and attack surface. The gateway centralises AML/velocity checks before traffic even reaches business logic. Spring Cloud Gateway on reactive Netty handles 50,000 RPS on 4 pods with non-blocking I/O.

Domain Entities

Entity: GatewayRoute

Defines a routing rule mapping an incoming URL path pattern to an upstream microservice. Stored in Redis for sub-millisecond lookup. Hot-reloaded from Spring Cloud Config without pod restarts.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
id	UUID primary key. Uniquely identifies this route configuration. System-generated.	f3a1c2d4-e5f6-...	UUID, PK, immutable
path_pattern	URL pattern to match incoming requests. Supports Ant-style wildcards. Unique — no two routes can match the same pattern.	/api/v1/payments/**	String, unique, not null
upstream_url	Internal Kubernetes service URL this route proxies to. Never exposed externally.	http://payment-svc:8085	String, not null
method_whitelist	Comma-separated HTTP methods allowed. Requests with other methods receive 405. Reduces attack surface.	GET,POST,PUT	String, default: all methods
rate_limit_rps	Max requests per second per client_id. Enforced atomically via Redis Lua script. 0 = unlimited for internal routes.	100	Integer, default: 100
auth_required	Whether JWT validation is mandatory. False only for public endpoints like /health, /emi-calculator.	true	Boolean, default: true

Entity: RateLimit

Tracks current request count for a client within a sliding time window. Stored in Redis with TTL so it auto-expires. Prevents API abuse and DDoS amplification.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
client_id	Identifies the rate-limited entity — customer UUID, partner API key hash, or IP for unauthenticated traffic.	partner-abc-001	String, Redis key component
window_seconds	Rolling window duration in seconds over which max_requests is counted.	60	Integer, typically 1, 10, or 60
max_requests	Maximum allowed requests within the window. Configurable per route and client tier.	100	Integer, not null
current_count	Atomic counter maintained by Redis INCR. Reset to 0 when window expires via TTL.	47	Integer, Redis TTL-managed, atomic
reset_at	Unix timestamp when the current window expires and count resets. Returned in X-RateLimit-Reset header.	1748900000	Long (epoch ms)

Entity: IPBlocklist

Blocklist of IP addresses denied at gateway perimeter. Dual-stored: Redis for hot sub-millisecond check; PostgreSQL for persistence and audit compliance.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
ip_address	IPv4 or IPv6 address to block. Supports CIDR notation for subnet-level blocks.	203.0.113.42	String(45), unique Redis key
reason	Human-readable reason: fraud, brute-force, scraping, DDoS. Required for compliance audit trail.	Brute force — 500 failed OTPs in 1hr	String, not null
blocked_at	Timestamp when block was applied. Used for SLA reporting.	2026-05-09T10:30:00Z	Timestamp with timezone, not null
blocked_by	SYSTEM for automated blocks; emp_id for manual security team actions.	SYSTEM	String, not null
expires_at	Null = permanent block. Non-null = auto-unblocked via Redis TTL.	2026-05-10T10:30:00Z	Timestamp, nullable

Entity: ApiKey

Credentials issued to partner banks and fintech partners for machine-to-machine API access. Full key never stored — only its SHA-256 hash.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
key_hash	SHA-256 hash of the actual API key. Original key given to partner once at issuance.	a3f1b2c4d5e6...	String(64), unique, indexed
partner_id	References the partner entity. Used for rate limiting, audit logs, and permission scoping.	partner-npci-001	String, FK → Partner
scopes	JSON array of allowed operations. Enforced by gateway filter before routing.	["READ_TXN","INITIATE_UP"]	JSON array, not null
created_at	Issuance timestamp. Used to calculate key age for rotation policy (max 1 year).	2026-01-15T09:00:00Z	Timestamp, not null
expires_at	Mandatory expiry — no indefinite keys. Maximum validity 1 year per security policy.	2027-01-15T09:00:00Z	Timestamp, not null
revoked	True if explicitly revoked before expiry. Checked before hash comparison to save CPU.	false	Boolean, default: false, indexed

Entity: CircuitBreakerState

Tracks Resilience4j circuit breaker state per upstream microservice. Stored in Redis so all gateway pods share state and don't independently flap.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
service_name	Internal Kubernetes service name. Acts as the circuit breaker key.	payment-service	String, Redis key
state	Current circuit state. CLOSED = normal. OPEN = failing, requests rejected fast. HALF_OPEN = testing recovery.	CLOSED	ENUM: CLOSED, OPEN, HALF_OPEN
failure_count	Number of failures in sliding window. Opens circuit when it exceeds 50% of calls in 10s.	3	Integer, Redis counter
last_failure_at	Timestamp of most recent failure. Used to calculate wait time before HALF_OPEN transition.	2026-05-09T14:22:11Z	Timestamp
next_retry_at	When circuit transitions from OPEN to HALF_OPEN for a probe request.	2026-05-09T14:22:41Z	Timestamp

REST API Endpoints

Method	Endpoint	Purpose
GET	/api/v1/health	Gateway health check — used by AWS ALB target group
GET	/api/v1/routes	List all active routes (admin only)
POST	/api/v1/routes	Register a new upstream route
DELETE	/api/v1/routes/{routeId}	Remove a route
GET	/api/v1/ratelimit/status/{clientId}	Check rate limit quota for a client
POST	/api/v1/blocklist/ip	Block an IP address (security team)
DELETE	/api/v1/blocklist/ip/{ip}	Unblock an IP address
GET	/api/v1/metrics	Prometheus metrics scrape endpoint

Method	Endpoint	Purpose
POST	/api/v1/apikeys	Provision API key for a partner
DELETE	/api/v1/apikeys/{keyId}	Revoke a partner API key

Kafka Topics

Direction	Topic Name	Trigger / Consumer
PUBLISHES	gateway.request.logged	Every inbound request — consumed by Audit Service
PUBLISHES	gateway.ratelimit.breached	Rate limit exceeded — consumed by Fraud Detection

Key Design Decisions

- GlobalFilter chain: JwtPreValidationFilter → RateLimitFilter → IPBlockFilter → RoutingFilter
- Redis Lua scripts ensure atomic rate-limit increment + expiry — no race conditions at high concurrency
- Circuit breaker per upstream service via Resilience4j — opens after 50% failure in 10s window
- All routes defined in YAML, hot-reloaded via Spring Cloud Config without pod restart
- mTLS enforced between gateway and all internal services — external traffic uses TLS 1.3 only

02 Identity & Authentication Service

OAuth2 · JWT · MFA · Session Management · RBAC

Purpose & Responsibility

Manages the complete identity lifecycle: customer registration verification, login (password + biometric), multi-factor authentication (OTP via SMS/TOTP app/hardware token), JWT access token + refresh token issuance, token introspection and revocation, role-based access control (RBAC) with banking-grade permission granularity (READ_BALANCE, INITIATE_TRANSFER, APPROVE_LOAN, etc.), and device fingerprinting to detect account takeover.

Architectural Rationale

Authentication is the most critical security service in a bank. Separating it means: (a) security patches deployed without touching business code, (b) auth service scaled independently during peak login hours (9–11 AM), (c) single authoritative token store prevents split-brain sessions. Spring Authorization Server 1.x implements full OAuth2/OIDC spec.

Domain Entities

Entity: User

Core identity record for every Nivesh Bank customer. Stores credentials and account security state. Never stores raw passwords — always bcrypt hashed. KYC status drives what operations the user can perform.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
user_id	UUID primary key. Immutable after creation. Shared as correlation ID across all microservices.	a1b2c3d4-e5f6-...	UUID, PK, immutable
mobile	10-digit mobile number. Primary login identifier. Verified via OTP during registration.	9876543210	String(10), unique, not null, indexed
email	Email address. Optional secondary identifier. Verified via email link during registration.	rohan@example.com	String, unique, nullable
password_hash	bcrypt hash with cost factor 12. Raw password never stored or logged anywhere.	\$2b\$12\$abc...xyz	String(60), bcrypt cost 12, not null
failed_attempts	Count of consecutive failed login attempts. Reset to 0 on success. Account locked when this hits 5.	0	Integer, default: 0, not null
locked_until	Null = account active. Non-null = locked until this timestamp. Set to now+30min after 5 failures.	2026-05-09T15:00:00Z	Timestamp, nullable
created_at	Timestamp of account creation. Immutable audit field.	2026-01-01T10:00:00Z	Timestamp, not null, immutable
kyc_status	Drives what the user can do. Checked before allowing account opening or financial transactions.	ACTIVE	ENUM: PENDING, KYC_IN_PROGRESS, KYC_VERIFIED, ACTIVE, SUSPENDED, CLOSED

Entity: Role

Named set of permissions assigned to users. System roles (CUSTOMER, BRANCH_TELLER, RM, COMPLIANCE_OFFICER, ADMIN) are pre-seeded. Custom roles can be created by admin.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
role_id	UUID PK.	r1o2l3e4-...	UUID, PK
role_name	Unique short name used in Spring Security @PreAuthorize expressions.	BRANCH_TELLER	String, unique, not null
description	Human-readable description of what this role allows. Displayed in admin portal.	Branch teller — can process cash transactions	String, not null
is_system_role	True for pre-seeded system roles that cannot be deleted.	true	Boolean, default: false

Entity: Permission

Atomic permission representing the right to perform one action on one resource. Fine-grained enough to allow READ_BALANCE without INITIATE_TRANSFER.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
permission_id	UUID PK.	p1e2r3m4-...	UUID, PK

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
permission_code	Short code used in @PreAuthorize annotations. Maps to Spring Security authority string.	INITIATE_TRANSFER	String, unique, not null
resource	Which domain resource this permission applies to.	TRANSACTION	String e.g. ACCOUNT, TRANSACTION, LOAN, CARD
action	What can be done on the resource.	WRITE	ENUM: READ, WRITE, APPROVE, ADMIN
description	Human-readable description for admin portal display.	Initiate fund transfers up to daily limit	String, not null

Entity: RolePermission

Junction table linking roles to permissions (many-to-many). Changing a role's permissions takes effect on next token issuance.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
role_id	FK to Role.	role_uuid	UUID, FK → Role, PK component
permission_id	FK to Permission.	perm_uuid	UUID, FK → Permission, PK component

Entity: UserRole

Links users to their assigned roles. Audit trail of who assigned which role and when. One user can have multiple roles.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
user_id	FK to User.	user_uuid	UUID, FK → User, indexed
role_id	FK to Role.	role_uuid	UUID, FK → Role
assigned_at	Timestamp of role assignment. Used in audit trail.	2026-01-15T09:00:00Z	Timestamp, not null
assigned_by	Employee ID of admin who assigned the role. SYSTEM for automated assignments on registration.	emp_00123	String, not null

Entity: RefreshToken

Stores bcrypt hash of each issued refresh token. Enables targeted revocation (logout from one or all devices). Each device gets its own refresh token.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
token_id	UUID PK. Returned to client as reference for revocation calls.	t1o2k3e4-...	UUID, PK
user_id	FK to User. One user can have many active refresh tokens (one per device/session).	user_uuid	UUID, FK → User, indexed
token_hash	bcrypt hash of the 256-bit random token. Full token given to client once; only hash stored.	\$2b\$12\$...	String(60), bcrypt hash, indexed
device_id	FK to DeviceFingerprint. Links refresh token to specific device. Null for web sessions.	device_uuid	UUID, nullable FK → DeviceFingerprint
expires_at	30-day expiry. After this, token rejected and client must re-authenticate with credentials.	2026-06-08T10:00:00Z	Timestamp, not null, indexed
revoked	True if explicitly revoked via logout. Checked before bcrypt comparison to save CPU.	false	Boolean, default: false, indexed
ip_address	IP from which token was issued. Used by fraud detection to flag use from different IPs.	203.0.113.42	String(45), supports IPv6

Entity: MfaSecret

Stores encrypted TOTP secret and backup codes for two-factor authentication. One record per user. TOTP secret AES-256-GCM encrypted with key from Vault.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
user_id	FK to User. One-to-one. Primary key is also user_id.	user_uuid	UUID, PK = FK → User
totp_secret_encrypted	AES-256-GCM encrypted TOTP seed (RFC 6238). Decrypted only during TOTP verification.	AES_ENCRYPTED_BASE64	String, AES-256-GCM encrypted
backup_codes_hash	Array of 8 one-time backup codes, each SHA-256 hashed. Code deleted on first use.	["\$sha256\$...", ...]	String[], SHA-256 each, length: 8 initially

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
preferred_method	Customer's chosen MFA channel. Determines what challenge is sent at login.	TOTP	ENUM: SMS_OTP, TOTP, HARDWARE_TOKEN, BIOMETRIC

Entity: OtpLog

Tracks every OTP generated and its usage. Prevents replay attacks (used_at set on first use). Supports multiple OTP purposes: login, transaction auth, password reset.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
otp_id	UUID PK.	otp_uuid	UUID, PK
user_id	FK to User who this OTP was generated for.	user_uuid	UUID, FK → User, indexed
otp_hash	SHA-256 hash of the 6-digit OTP. Actual OTP sent to device; only hash stored.	sha256_hash_value	String(64), SHA-256
purpose	Why this OTP was generated. Enforces that LOGIN OTP cannot be used for TRANSACTION_AUTH.	LOGIN	ENUM: LOGIN, TRANSACTION_AUTH, PASSWORD_RESET, KYC_VERIFY, DEVICE_TRUST
expires_at	OTP valid for 5 minutes. Rejected if used after this timestamp.	2026-05-09T10:35:00Z	Timestamp, not null, indexed
used_at	Null if unused. Populated on first valid use. Prevents OTP reuse (replay attack prevention).	2026-05-09T10:32:00Z	Timestamp, nullable
channel	How the OTP was delivered to the customer.	SMS	ENUM: SMS, EMAIL, TOTP_APP, HARDWARE_TOKEN

Entity: DeviceFingerprint

Tracks customer devices to detect account takeover. Transaction from a new untrusted device triggers additional MFA.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
device_id	UUID PK. Persisted in device secure storage (iOS Keychain / Android Keystore).	device_uuid	UUID, PK
user_id	FK to User. One user can have many registered devices.	user_uuid	UUID, FK → User, indexed
fingerprint_hash	SHA-256 hash of device signals: IP subnet, user-agent, screen resolution, installed fonts.	sha256_fingerprint	String(64), SHA-256
platform	Operating system platform of the device.	ANDROID	ENUM: ANDROID, IOS, WEB, WINDOWS, MAC
trusted	True if customer explicitly marked this device as trusted after MFA.	true	Boolean, default: false
registered_at	Timestamp device was first seen and registered.	2026-01-10T09:00:00Z	Timestamp, not null

Entity: LoginAudit

Immutable log of every login attempt. Feeds fraud detection (failed login patterns), compliance (access audit), and customer support (last login). 10-year retention.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
log_id	UUID PK. Append-only — no updates or deletes. 10-year retention per RBI IT framework.	log_uuid	UUID, PK, append-only
user_id	Which user attempted login. Indexed for quick retrieval of login history.	user_uuid	UUID, FK → User, indexed
ip	Client IP address. Used for geolocation-based anomaly detection.	203.0.113.42	String(45)
device_id	Device fingerprint at time of attempt. Null if device not previously registered.	device_uuid	UUID, nullable
success	True if login completed including MFA. False for wrong password, OTP timeout, locked account.	true	Boolean, not null
failure_reason	Null on success. Specific reason code on failure for fraud velocity analysis.	INVALID_PASSWORD	ENUM: INVALID_PASSWORD, OTP_EXPIRED, ACCOUNT_LOCKED, DEVICE_MISMATCH, null

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
timestamp	Exact server-side timestamp. Used for velocity checks (N failures in M minutes).	2026-05-09T10:00:00.123Z	Timestamp(6), not null, indexed

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/auth/register	New customer self-registration with KYC initiation
POST	/api/v1/auth/login	Credential login — returns challenge if MFA required
POST	/api/v1/auth/mfa/verify	Verify OTP or TOTP code — issues access + refresh tokens
POST	/api/v1/auth/token/refresh	Exchange valid refresh token for new access token
POST	/api/v1/auth/token/revoke	Revoke refresh token (logout)
POST	/api/v1/auth/token/introspect	Validate token and return claims (called by Gateway)
POST	/api/v1/auth/otp/send	Send OTP via SMS/email for a given purpose
POST	/api/v1/auth/otp/verify	Verify OTP without full login (e.g. transaction auth)
POST	/api/v1/auth/password/change	Change password (requires current password + OTP)
POST	/api/v1/auth/password/reset	Forgot-password flow via verified mobile/email
POST	/api/v1/auth/biometric/register	Register biometric public key for a device
POST	/api/v1/auth/biometric/verify	Login via biometric challenge-response
POST	/api/v1/auth/mfa/setup/totp	Enrol Google Authenticator — returns QR code
POST	/api/v1/auth/device/trust	Mark device as trusted after MFA
GET	/api/v1/auth/sessions	List all active sessions for authenticated user
DELETE	/api/v1/auth/sessions/{id}	Terminate a specific session
GET	/api/v1/admin/roles	List all system roles (admin)
POST	/api/v1/admin/roles	Create a custom role
POST	/api/v1/admin/users/{id}/roles	Assign role to user
POST	/api/v1/admin/users/{id}/lock	Lock a user account (compliance action)

Kafka Topics

Direction	Topic Name	Trigger / Consumer
PUBLISHES	auth.user.registered	Customer Profile Service — triggers profile creation
PUBLISHES	auth.login.failed	Fraud Detection — triggers anomaly scoring
PUBLISHES	auth.account.locked	Notification Service — alert customer
PUBLISHES	auth.password.changed	Notification Service + Audit — security event

Key Design Decisions

- JWT access tokens have 15-minute expiry; refresh tokens 30-day expiry stored as bcrypt hash
- Failed login counter uses Redis atomic INCR with TTL — account locked after 5 failures in 30 min
- TOTP uses RFC 6238 standard; backup codes are 8-character one-time codes hashed with SHA-256
- Device fingerprinting compares: IP subnet, user-agent, screen resolution, installed fonts hash
- All password operations use bcrypt cost factor 12 — rotated to Argon2id in v2
- Spring Security filter chain: JwtFilter → MfaFilter → RateLimitFilter → AuthorizationFilter

03 Customer Profile Service

KYC · Demographics · Documents · Relationship Management

Purpose & Responsibility

The single source of truth for all customer demographic and KYC data. Manages personal information (name, DOB, PAN, Aadhaar, passport), KYC document upload and verification (via UIDAI Aadhaar API, NSDL PAN verification API, video KYC), nominee management, address management (multiple addresses — permanent, current, correspondence), customer segmentation (Retail/HNI/Corporate/NRI), relationship manager assignment, and customer risk profiling.

Architectural Rationale

Customer data is the most sensitive PII in a bank. A dedicated service allows: strict data access policies, independent GDPR/DPDPA compliance controls, field-level encryption of Aadhaar/PAN, and the ability to deploy a separate data residency zone if regulatory requirements change. MongoDB stores unstructured document metadata alongside relational PostgreSQL for structured profile data.

Domain Entities

Entity: Customer

Master customer record. Created immediately after `auth.user.registered` event. Starts in `PENDING_KYC` and progresses through the KYC state machine before any financial product can be opened.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
<code>customer_id</code>	UUID PK. The bank-wide customer identifier used in all service-to-service calls.	<code>c1d2e3f4-...</code>	UUID, PK, immutable
<code>user_id</code>	FK to Auth Service User. One-to-one. Links identity credentials to customer profile.	<code>user_uuid</code>	UUID, FK, unique, not null
<code>first_name</code>	Legal first name as per KYC documents. Updated only via verified KYC process — not self-service after verification.	<code>Rohan</code>	String, not null, max 50 chars
<code>last_name</code>	Legal last name as per KYC documents.	<code>Sharma</code>	String, not null, max 50 chars
<code>dob</code>	Date of birth. Used for age verification (18+ for account opening), nominee minor check, and birthday rewards.	<code>1996-04-15</code>	Date, not null
<code>gender</code>	Customer's gender as declared during registration.	<code>MALE</code>	ENUM: MALE, FEMALE, OTHER, PREFER_NOT_TO_SAY
<code>nationality</code>	Customer's nationality. Used for NRI account eligibility and FATCA classification.	<code>INDIAN</code>	String(3), ISO 3166-1 alpha-3
<code>segment</code>	Customer segment determines product eligibility, interest rates, and service levels. Auto-upgraded by batch job when criteria met.	<code>RETAIL</code>	ENUM: RETAIL, HNI, CORPORATE, NRI, JAN_DHAN
<code>risk_profile</code>	Risk classification for AML and investment suitability. Set by compliance during KYC review.	<code>MEDIUM</code>	ENUM: LOW, MEDIUM, HIGH, VERY_HIGH
<code>rm_id</code>	Assigned Relationship Manager. Null for Retail segment. Mandatory for HNI and Corporate.	<code>rm_uuid</code>	UUID, nullable FK → RelationshipManager
<code>created_at</code>	Profile creation timestamp. Immutable.	<code>2026-01-01T10:00:00Z</code>	Timestamp, not null, immutable
<code>status</code>	Drives what the customer can do. <code>PENDING_KYC</code> cannot open accounts. <code>SUSPENDED</code> freezes all transactions.	<code>ACTIVE</code>	ENUM: PENDING_KYC, KYC_IN_PROGRESS, KYC_VERIFIED, ACTIVE, SUSPENDED, CLOSED

Entity: CustomerAddress

Stores multiple addresses for a customer. At least one `PERMANENT` address required before account opening. Address verification done via postal API or physical document.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
<code>address_id</code>	UUID PK.	<code>addr_uuid</code>	UUID, PK
<code>customer_id</code>	FK to Customer. One customer can have multiple addresses.	<code>cust_uuid</code>	UUID, FK → Customer, indexed

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
type	Address category. PERMANENT is the legal address for KYC. CORRESPONDENCE used for mailings.	PERMANENT	ENUM: PERMANENT, CURRENT, CORRESPONDENCE
address_lines	Full address text split into lines. Line 1 mandatory; Lines 2-3 optional.	Flat 4B, Orchid Heights	String[], not null (line1), nullable (line2, line3)
city	City name.	Pune	String, not null
state	State name.	Maharashtra	String, not null
pincode	6-digit PIN code. Used for IFSC branch assignment and fraud geo-analysis.	411001	String(6), not null
country	Country code ISO 3166-1 alpha-3. Non-IND for NRI customers.	IND	String(3), not null, default: IND
verified	True once verified via Aadhaar OTP or physical document check. Unverified addresses cannot be used for KYC.	true	Boolean, default: false
default_flag	True for the primary address. Only one address per type can be default.	true	Boolean, default: false

Entity: CustomerContact

Stores verified contact details. PRIMARY mobile and email used for all OTP and alerts. SECONDARY contacts used for backup communication.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
contact_id	UUID PK.	contact_uuid	UUID, PK
customer_id	FK to Customer.	cust_uuid	UUID, FK → Customer, indexed
mobile	Mobile number. Must be unique across all customers. Verified via OTP before activation.	9876543210	String(10), unique when type=PRIMARY
email	Email address. Verified via email link before activation.	rohan@example.com	String, nullable
type	Whether this is the primary or secondary contact.	PRIMARY	ENUM: PRIMARY, SECONDARY
verified	True only after OTP or email link verification.	true	Boolean, default: false
updated_at	Timestamp of last update. Used to detect suspicious rapid contact changes.	2026-05-01T09:00:00Z	Timestamp, not null

Entity: KycDocument

Represents one uploaded KYC document. Each goes through a verification workflow. Actual file in S3 with SSE-KMS encryption; only metadata in DB. Document number AES-256-GCM encrypted.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
doc_id	UUID PK.	doc_uuid	UUID, PK
customer_id	FK to Customer. One customer can have multiple documents of different types.	cust_uuid	UUID, FK → Customer, indexed
doc_type	Which identity document this is. Determines which verification API is called.	AADHAAR	ENUM: PAN, AADHAAR, PASSPORT, VOTER_ID, DRIVING_LICENSE
doc_number_encrypted	AES-256-GCM encrypted document number. Key managed by HashiCorp Vault. Never stored in plaintext.	AES_ENC_BASE64	String, AES-256-GCM encrypted
file_path	S3 object key for the uploaded document. SSE-KMS encrypted at rest.	s3://nvsh-kyc/c1d2/aadhaar_front.jpg	String, S3 path, not null
verified_at	Timestamp when verification API confirmed the document. Null if pending or failed.	2026-05-09T11:00:00Z	Timestamp, nullable
verified_by	Who verified: UIDAI_API, NSDL_API, VIDEO_KYC_AGENT, or emp_id for manual.	UIDAI_API	String — ENUM value or employee_id
expiry_date	Document expiry. Null for PAN. System triggers re-KYC before expiry for passport/DL.	2035-01-01	Date, nullable
status	Current state of document verification.	VERIFIED	ENUM: UPLOADED, UNDER_REVIEW, VERIFIED, REJECTED, EXPIRED

Entity: Nominee

A nominee receives account proceeds upon the account holder's death. Mandatory for new account opening per RBI guidelines. Minor nominees require a guardian.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
nominee_id	UUID PK.	nom_uuid	UUID, PK
customer_id	FK to Customer — the account holder who added this nominee.	cust_uuid	UUID, FK → Customer
name	Full legal name of the nominee.	Priya Sharma	String, not null
relation	Relationship of nominee to account holder. Required by RBI nomination guidelines.	SPOUSE	ENUM: SPOUSE, SON, DAUGHTER, FATHER, MOTHER, SIBLING, OTHER
dob	Nominee's date of birth. Determines if nominee is a minor (under 18) requiring guardian details.	1998-07-20	Date, not null
share_percent	Percentage of proceeds allocated. Multiple nominees must sum to exactly 100%.	100.00	DECIMAL(5,2), check: 0 < share <= 100
address_id	FK to CustomerAddress. Nominee's address for legal communication.	addr_uuid	UUID, nullable FK → CustomerAddress
guardian_name	Required only if nominee is a minor. Guardian manages funds until nominee turns 18.	Rahul Sharma	String, nullable — non-null only if dob indicates minor

Entity: CustomerSegment

Defines criteria and benefits for each customer segment. Used by segment upgrade batch job to auto-upgrade customers when criteria met.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
segment_id	UUID PK.	seg_uuid	UUID, PK
name	Segment name matching the ENUM in Customer.segment.	HNI	ENUM: RETAIL, HNI, CORPORATE, NRI, JAN_DHAN
min_balance	Minimum average quarterly balance required to stay in this segment.	1000000.0000	DECIMAL(20,4)
benefits_json	JSON describing benefits: free RTGS, dedicated RM, lounge access, etc.	{"free_rtgs":true,"rm_assigned":true}	JSON, not null
upgrade_criteria	Human-readable criteria used in marketing and customer communications.	AQB >= 10L for 3 consecutive quarters	String, not null

Entity: RelationshipManager

Bank employee assigned to manage HNI/Corporate customer relationships. Tracks capacity to prevent over-allocation.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
rm_id	UUID PK.	rm_uuid	UUID, PK
employee_id	Bank's HR system employee ID. Used for internal cross-referencing.	EMP00456	String, unique, not null
name	RM's full name displayed to customers.	Aditya Mehta	String, not null
branch_id	Branch where the RM is based.	branch_uuid	UUID, FK → Branch
max_customers	Maximum customers this RM can manage. Enforced at assignment time.	50	Integer, not null
specialization	Domain specialization for routing complex customer needs.	WEALTH_MANAGEMENT	ENUM: RETAIL_LOANS, WEALTH_MANAGEMENT, CORPORATE_BANKING, NRI_SERVICES

Entity: VideoKYC

Tracks a video KYC session where customer completes KYC via live video call with a bank agent. Recording stored in S3 for audit. Stored in MongoDB.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
vkyc_id	UUID PK.	vkyc_uuid	UUID, PK (MongoDB_id)
customer_id	FK to Customer undergoing video KYC.	cust_uuid	UUID, indexed

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
session_url	One-time video session URL. Expires after 30 minutes if not joined.	https://vkyc.nivesh.bank/session/abc	String, TTL-based
agent_id	Bank employee ID of the video KYC agent who conducted the session.	EMP00789	String, nullable — null until agent joins
status	Session lifecycle state.	COMPLETED	ENUM: SCHEDULED, IN_PROGRESS, COMPLETED, FAILED, CANCELLED
completed_at	Timestamp when session successfully completed. Null if not yet done.	2026-05-09T14:00:00Z	Timestamp, nullable
recording_s3_path	S3 path to the encrypted session recording. Required for audit and disputes.	s3://nvsh-vkyc/recordings/session_abc.mp4	String, nullable — populated after completion

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/customers	Create customer profile (triggered after auth registration)
GET	/api/v1/customers/{customerId}	Fetch full customer profile
PUT	/api/v1/customers/{customerId}	Update personal details (name, contact, etc.)
GET	/api/v1/customers/{customerId}/addresses	List all addresses
POST	/api/v1/customers/{customerId}/addresses	Add new address
PUT	/api/v1/customers/{customerId}/addresses/{addressId}	Update address
POST	/api/v1/customers/{customerId}/kyc/documents	Upload KYC document to S3 + trigger verification
GET	/api/v1/customers/{customerId}/kyc/status	Get overall KYC completion status
POST	/api/v1/customers/{customerId}/kyc/aadhaar/verify	Trigger UIDAI OTP-based Aadhaar verification
POST	/api/v1/customers/{customerId}/kyc/pan/verify	Trigger NSDL PAN verification
POST	/api/v1/customers/{customerId}/kyc/video	Initiate video KYC session
GET	/api/v1/customers/{customerId}/nominees	List nominees
POST	/api/v1/customers/{customerId}/nominees	Add nominee with share percentage
PUT	/api/v1/customers/{customerId}/nominees/{nomineeId}	Update nominee details
DELETE	/api/v1/customers/{customerId}/nominees/{nomineeId}	Remove nominee
GET	/api/v1/customers/search?pan=&mobile;=	Search customer by PAN/mobile (staff only)
POST	/api/v1/customers/{customerId}/segment/upgrade	Trigger segment upgrade evaluation
GET	/api/v1/customers/{customerId}/risk-profile	Get customer risk assessment
PUT	/api/v1/customers/{customerId}/rm	Assign/change Relationship Manager

Kafka Topics

Direction	Topic Name	Trigger / Consumer
PUBLISHES	customer.kyc.completed	Account Service — enables account opening
PUBLISHES	customer.profile.updated	Notification + Audit Service
PUBLISHES	customer.risk.changed	Fraud Detection Service
SUBSCRIBES	auth.user.registered	Creates initial customer record

Key Design Decisions

- Aadhaar number stored as AES-256-GCM encrypted — key in HashiCorp Vault, rotated annually
- KYC documents uploaded to S3 with SSE-KMS; pre-signed URLs expire in 15 min
- Customer status machine: PENDING_KYC → KYC_IN_PROGRESS → KYC_VERIFIED → ACTIVE → SUSPENDED → CLOSED
- PAN/Aadhaar uniqueness enforced via encrypted hash index — prevents duplicate customer creation
- Re-KYC workflow triggered automatically every 2 years per RBI guidelines via scheduled job

04

Account Management Service

Account lifecycle · Balance · Interest · Dormancy

Purpose & Responsibility

Manages the lifecycle of all bank accounts: Savings, Current, Fixed Deposit (FD), Recurring Deposit (RD), NRI accounts (NRE/NRO/FCNR), Salary accounts, Jan Dhan accounts, and PPF accounts. Handles account opening (post-KYC), IFSC/account number generation, balance enquiry, interest calculation (daily accrual, monthly credit), minimum balance enforcement, dormancy detection, account statements, and account closure workflows.

Architectural Rationale

Account data is the financial backbone of the bank. Separating it from transactions allows: independent scaling of balance reads (extremely frequent) vs. balance updates (transactional). The Account Service owns the canonical balance and acts as the ledger authority. Transaction Service debits/credits accounts via an internal API with idempotency guarantees — never directly manipulating the accounts table.

Domain Entities

Entity: Account

Master record for a bank account. Each account has an 11-digit account number (Type+Region+Sequence+Luhn), belongs to one customer, and maintains DECIMAL(20,4) balance for cent-exact precision. Status drives what operations are permitted.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
account_id	UUID internal PK. Used in all service-to-service API calls. Not shown to customers.	acc_uuid_123	UUID, PK, immutable
account_number	11-digit account number per Nivesh Bank format: Type(1)+Region(3)+Sequence(6)+Luhn(1). Luhn-validated. Unique globally. What customers see and share.	14000000017	String(11), unique, indexed, Luhn-validated
customer_id	FK to Customer. One customer can have multiple accounts (savings + FD + loan, etc.).	cust_uuid	UUID, FK → Customer, indexed
account_type	Type of account — determines applicable interest rates, limits, and regulatory treatment.	SAVINGS	ENUM: SAVINGS, CURRENT, FD, RD, NRE, NRO, FCNR, SALARY, JAN_DHAN, PPF
ifsc_code	11-char IFSC of the branch where account was opened. Format: NVSH0XXXXXX.	NVSH0400001	String(11), FK → Branch.ifsc_code
branch_id	FK to Branch where account was opened. Used for dormancy and legal notices.	branch_uuid	UUID, FK → Branch
currency	Account currency. INR for domestic; USD/GBP/EUR for FCNR accounts.	INR	String(3), ISO 4217, default: INR
balance	Current ledger balance. DECIMAL(20,4) — never Float or Double. Supports balances up to 9,999,999,999,999,999.9999.	125000.5000	DECIMAL(20,4), not null, check >= 0 for savings
available_balance	Balance minus any holds (cheques in clearing, RTGS in progress). What the customer can actually spend.	124500.0000	DECIMAL(20,4), derived, cached in Redis 5s TTL
status	Controls what operations are permitted. FROZEN blocks all debits. DORMANT requires branch reactivation. CLOSED is terminal.	ACTIVE	ENUM: ACTIVE, FROZEN, DORMANT, CLOSED, PENDING_ACTIVATION
opened_date	Date account was activated. After this date, interest accrual begins.	2026-01-15	Date, not null, immutable
maturity_date	Applicable to FD and RD only. Null for savings/current. Triggers auto-credit on maturity.	2027-01-15	Date, nullable — non-null only for FD/RD
interest_rate	Annual interest rate. For savings: RBI-linked. For FD: fixed at opening. For current: 0.	3.50	DECIMAL(5,2), percent per annum
nomination_id	FK to Nominee. The nominee for this specific account.	nom_uuid	UUID, nullable FK → Nominee

Entity: AccountType

Configuration table defining rules for each account type. Drives minimum balance, interest rates, withdrawal limits, and feature flags. Updated by product team.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
type_id	UUID PK.	type_uuid	UUID, PK
name	Account type name matching Account.account_type ENUM.	SAVINGS	ENUM value, unique
min_balance	Minimum average monthly balance required. Penalty applied if breached.	10000.0000	DECIMAL(20,4), 0 for JAN_DHAN
interest_rate	Default annual interest rate for this account type.	3.50	DECIMAL(5,2)
max_withdrawal_limit	Maximum per-transaction withdrawal limit in INR.	100000.0000	DECIMAL(20,4)
features_json	JSON array of feature flags: internet_banking, upi, cheque_book, etc.	{"upi":true,"cheque_book":true}	JSON, not null

Entity: FixedDeposit

Details of a Fixed Deposit linked to a savings/current account. FDs earn higher interest. Premature withdrawal incurs penalty. Auto-renewal renews at prevailing rates on maturity.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
fd_id	UUID PK.	fd_uuid	UUID, PK
account_id	FK to parent savings account that funded and will receive maturity proceeds.	acc_uuid	UUID, FK → Account, indexed
principal	Amount invested. Locked for tenure. Minimum INR 1,000.	100000.0000	DECIMAL(20,4), check >= 1000
tenure_months	Duration of FD in months. Ranges from 1 to 120.	12	Integer, not null
interest_rate	Fixed rate at time of FD opening. Does not change during tenure.	7.25	DECIMAL(5,2), locked at creation
maturity_amount	Pre-calculated maturity value at FD opening.	107250.0000	DECIMAL(20,4)
payout_type	CUMULATIVE: compounded and paid at maturity. MONTHLY/QUARTERLY: paid to linked account periodically.	CUMULATIVE	ENUM: CUMULATIVE, MONTHLY, QUARTERLY
auto_renewal	If true, FD auto-renews at prevailing rates on maturity for same tenure.	true	Boolean, default: true
maturity_date	Date FD matures. account.fd.matured event published on this date.	2027-01-15	Date, not null

Entity: RecurringDeposit

Details of a Recurring Deposit — customer deposits a fixed amount every month. Missed instalments incur penalty. Maturity amount calculated at opening using compound interest.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
rd_id	UUID PK.	rd_uuid	UUID, PK
account_id	FK to parent savings account. Monthly debit happens from this account.	acc_uuid	UUID, FK → Account, indexed
monthly_instalment	Fixed amount debited from linked savings account each month.	5000.0000	DECIMAL(20,4), check > 0
tenure_months	Total duration of RD in months.	24	Integer, not null
interest_rate	Fixed interest rate locked at RD opening.	6.50	DECIMAL(5,2)
maturity_amount	Pre-calculated maturity value assuming all instalments paid on time.	128700.0000	DECIMAL(20,4)
next_debit_date	Date of next monthly instalment debit. Updated after each debit.	2026-06-15	Date, not null
missed_count	Number of missed instalments. After 3 consecutive misses, RD is considered broken.	0	Integer, default: 0

Entity: InterestAccrual

Daily interest accrual record created by Spring Batch nightly job at 11:59 PM. Accumulates accrued (not yet posted) interest. Posted=true when monthly credit happens.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
accrual_id	UUID PK.	accrual_uuid	UUID, PK
account_id	FK to Account for which interest was accrued.	acc_uuid	UUID, FK → Account, indexed
accrual_date	Date for which interest was accrued. One record per account per day.	2026-05-09	Date, unique with account_id
accrued_amount	Daily interest: (balance * rate) / 365.	10.2740	DECIMAL(20,4), not null
rate_applied	Interest rate used for this day's calculation. Captures rate changes mid-period.	3.50	DECIMAL(5,2)
posted	False until month-end when accrued amounts are aggregated and credited to the account.	false	Boolean, default: false, indexed

Entity: MinBalancePenalty

Record of minimum balance shortfall penalty charges. Created when account balance falls below required minimum. Can be waived by branch manager.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
penalty_id	UUID PK.	penalty_uuid	UUID, PK
account_id	FK to Account that incurred the penalty.	acc_uuid	UUID, FK → Account
charge_date	Date the penalty was assessed.	2026-05-01	Date, not null
shortfall_amount	How much below minimum balance the account was.	3000.0000	DECIMAL(20,4), not null
charge_amount	Penalty fee charged. Typically INR 500 + 18% GST.	590.0000	DECIMAL(20,4), not null
waived	True if branch manager waived the penalty. Once waived, cannot be reversed.	false	Boolean, default: false

Entity: DormancyRecord

Tracks when account was declared dormant (no transactions for 2 years per RBI). After 10 years, balance transferred to RBI Depositor Education and Awareness Fund (DEAF).

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
dormancy_id	UUID PK.	dorm_uuid	UUID, PK
account_id	FK to Account. One dormancy record per dormancy episode.	acc_uuid	UUID, FK → Account, indexed
declared_dormant_at	Timestamp when account moved to DORMANT status by nightly batch job.	2026-03-01T00:01:00Z	Timestamp, not null
last_txn_date	Date of last customer-initiated transaction before dormancy. Used for 2-year and 10-year threshold calculations.	2024-02-28	Date, not null
reactivated_at	Null if still dormant. Populated when customer visits branch with ID proof for reactivation.	2026-05-01T14:30:00Z	Timestamp, nullable

Entity: AccountBeneficiary

Saved beneficiary for fund transfers. Verified beneficiaries can receive transfers without additional confirmation. Verified by making a small test credit (penny drop).

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
beneficiary_id	UUID PK.	ben_uuid	UUID, PK
account_id	FK to Account of the customer who added this beneficiary.	acc_uuid	UUID, FK → Account, indexed
name	Beneficiary's full name. Shown in transfer confirmation screen.	Priya Sharma	String, not null
ifsc	Beneficiary bank's IFSC code. Validated against RBI IFSC master list.	HDFC0001234	String(11), not null
account_number	Beneficiary's account number at their bank.	12345678901	String, not null
nickname	Customer-assigned nickname for quick identification.	Wife	String, nullable
verified	True once penny-drop verification credit confirmed the account exists.	true	Boolean, default: false

Entity: JointHolder

Tracks additional account holders for joint accounts. Operation mode defines whether both signatures needed or either holder can operate independently.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
holder_id	UUID PK.	holder_uuid	UUID, PK
account_id	FK to Account.	acc_uuid	UUID, FK → Account
customer_id	FK to Customer who is the joint holder. Must have KYC_VERIFIED status.	cust_uuid	UUID, FK → Customer
operation_mode	EITHER_OR: any one holder can operate alone. JOINTLY: both signatures required for debits.	EITHER_OR	ENUM: EITHER_OR, JOINTLY
added_at	Timestamp when joint holder was added. Requires both holders' consent and branch verification.	2026-02-01T10:00:00Z	Timestamp, not null

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/accounts	Open new account (requires KYC_VERIFIED status)
GET	/api/v1/accounts/{accountId}	Get account details and current balance
GET	/api/v1/accounts?customerId={id}	List all accounts for a customer
GET	/api/v1/accounts/{accountId}/balance	Real-time balance (Redis cache → DB fallback, 5s TTL)
GET	/api/v1/accounts/{accountId}/mini-statement	Last 10 transactions summary
PUT	/api/v1/accounts/{accountId}/status	Freeze/unfreeze/close account
POST	/api/v1/accounts/{accountId}/fd	Open Fixed Deposit linked to savings account
GET	/api/v1/accounts/{accountId}/fd/{fdId}	FD details including maturity projection
PUT	/api/v1/accounts/{accountId}/fd/{fdId}/renew	Set/change auto-renewal preference
POST	/api/v1/accounts/{accountId}/rd	Open Recurring Deposit
GET	/api/v1/accounts/{accountId}/rd/{rdId}	RD status and payment schedule
GET	/api/v1/accounts/{accountId}/interest/accruals	Interest accrual history
POST	/api/v1/accounts/{accountId}/beneficiaries	Add fund transfer beneficiary
GET	/api/v1/accounts/{accountId}/beneficiaries	List registered beneficiaries
DELETE	/api/v1/accounts/{accountId}/beneficiaries/{benId}	Remove beneficiary
POST	/api/v1/accounts/{accountId}/joint-holders	Add joint account holder
GET	/api/v1/accounts/ifsc/{ifscCode}	Lookup branch details by IFSC
PUT	/api/v1/accounts/{accountId}/nominee	Update account nominee
POST	/api/v1/internal/accounts/{accountId}/debit	INTERNAL: Debit account (Transaction Svc only)
POST	/api/v1/internal/accounts/{accountId}/credit	INTERNAL: Credit account (Transaction Svc only)

Kafka Topics

Direction	Topic Name	Trigger / Consumer
PUBLISHES	account.opened	Notification Service — welcome message
PUBLISHES	account.balance.low	Notification Service — low balance alert
PUBLISHES	account.dormant	Notification + Compliance Service
PUBLISHES	account.fd.matured	Transaction Service — trigger auto-credit of maturity amount
SUBSCRIBES	customer.kyc.completed	Pre-validates KYC before account marked ACTIVE
SUBSCRIBES	transaction.completed	Updates cached balance in Redis

Key Design Decisions

- Account number: 11-digit numeric — Type(1)+Region(3)+Sequence(6)+Luhn check digit(1)

- Balance stored in PostgreSQL DECIMAL(20,4) — never Float/Double — for cent-exact precision
- Redis cache stores current_balance per account_id with 5-second TTL for balance enquiry scalability
- Interest calculated via daily accrual batch job (Spring Batch) running at 11:59 PM — credits monthly
- Dormancy: no transaction for 2 years → DORMANT; 10 years → transfer to RBI Depositor Education Fund
- Optimistic locking (@Version) on Account entity prevents concurrent balance corruption

05 Transaction Processing Service

Core ledger · Double-entry bookkeeping · ATM · POS

Purpose & Responsibility

The heart of the banking platform — processes every financial movement: cash deposits, cash withdrawals, ATM transactions, POS/merchant transactions, internal account transfers, fee debits, interest credits, cheque deposits, DD issuance, and standing instruction executions. Implements double-entry bookkeeping (every debit has a corresponding credit). Cassandra stores transaction history for time-series retrieval at massive scale.

Architectural Rationale

Transaction processing is the highest-throughput, most failure-sensitive operation in a bank. By isolating it, we can tune the JVM heap for high-concurrency (virtual threads in Java 21), deploy on I/O-optimised instances, apply separate disaster recovery, and maintain a clean audit trail independently. Cassandra's time-series model serves 5-year transaction history queries in milliseconds.

Domain Entities

Entity: Transaction

Master transaction record. Every debit or credit creates one Transaction row and exactly two JournalEntry rows (double-entry). Hot data in PostgreSQL (current month); history >90 days moves to Cassandra.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
txn_id	UUID PK. Unique across all transactions ever. Returned to client for status polling.	txn_uuid	UUID, PK, immutable, indexed
reference_number	Human-readable reference visible on statements and SMS. Format: YYYYMMDD + 8-digit seq.	20260509000012345	String(18), unique, indexed
type	Direction and nature of the transaction.	DEBIT	ENUM: DEBIT, CREDIT, REVERSAL, FEE, INTEREST, REFUND
debit_account_id	FK to Account being debited. Null for inbound credits from external banks.	acc_uuid	UUID, FK → Account, nullable
credit_account_id	FK to Account being credited. Null for outbound payments to external banks.	acc_uuid	UUID, FK → Account, nullable
amount	Transaction amount. DECIMAL(20,4). Must be > 0.	5000.0000	DECIMAL(20,4), check > 0, not null
currency	ISO 4217 currency code.	INR	String(3), default: INR
status	State machine. Terminal states: POSTED (success), FAILED, REVERSED, BLOCKED.	POSTED	ENUM: INITIATED, FRAUD_CHECK, PENDING, POSTED, FAILED, REVERSED, BLOCKED
channel	How the transaction was initiated. Used for limit validation and analytics.	UPI	ENUM: BRANCH, ATM, INTERNET_BANKING, MOBILE_APP, UPI, NEFT, RTGS, IMPS, API
description	Narration for the transaction shown on statement.	UPI Transfer to Priya	String(200), nullable
created_at	Server-side timestamp when initiated. Immutable. Partitioned by month in PostgreSQL.	2026-05-09T10:30:00.123Z	Timestamp(6), not null, partitioned by month
settled_at	Timestamp when transaction was fully settled. Null until POSTED.	2026-05-09T10:30:02Z	Timestamp, nullable
reversal_txn_id	FK to the reversal Transaction row. Null for normal transactions.	reversal_uuid	UUID, nullable FK → Transaction

Entity: TransactionType

Configuration table defining rules for each transaction type. Controls daily limits, whether charges apply, and whether reversals are allowed.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
type_code	Short code matching Transaction.type ENUM.	UPI_TRANSFER	String, PK, unique
name	Human-readable name shown on statements.	UPI Fund Transfer	String, not null

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
gl_code	General Ledger account code for regulatory reporting.	GL-2001-UPI	String, FK → GLAccount
max_limit_daily	Maximum daily amount allowed for this transaction type per account.	100000.0000	DECIMAL(20,4)
charge_applicable	Whether a service charge is levied for this transaction type.	false	Boolean, not null
reversal_allowed	Whether this transaction type can be reversed.	true	Boolean, not null

Entity: JournalEntry

Implements double-entry bookkeeping. Every Transaction creates exactly 2 JournalEntry rows: one DR and one CR. DB constraint enforces this. The ledger never has single-entry transactions.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
entry_id	UUID PK.	entry_uuid	UUID, PK
txn_id	FK to Transaction. Each transaction has exactly 2 entries — enforced by DB CHECK constraint.	txn_uuid	UUID, FK → Transaction, indexed
account_id	Which account this entry affects.	acc_uuid	UUID, FK → Account
dr_cr	Whether this entry debits or credits the account.	DR	ENUM: DR (Debit reduces balance), CR (Credit increases balance)
amount	Entry amount. Must equal the parent transaction amount. DR and CR must be equal.	5000.0000	DECIMAL(20,4), check > 0
gl_account	General Ledger account code for RBI regulatory reporting.	GL-1001-SAVINGS	String, FK → GLAccount
entry_date	Date of the journal entry. Used for day-wise GL reporting.	2026-05-09	Date, not null
narration	Human-readable description for the ledger entry.	UPI Transfer to Priya via BHIM	String(200)

Entity: AtmTransaction

Extended details for ATM-originated transactions. Tracks ATM-specific details: PIN verification, dispensed amount (may differ from requested if ATM ran low on cash).

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
atm_txn_id	UUID PK.	atm_txn_uuid	UUID, PK
txn_id	FK to parent Transaction.	txn_uuid	UUID, FK → Transaction
atm_id	FK to AtmTerminal. Identifies which physical ATM processed this transaction.	atm_uuid	UUID, FK → AtmTerminal
card_id	FK to Card used at the ATM.	card_uuid	UUID, FK → Card
pin_verified	True if cardholder entered correct PIN. False transactions flagged for fraud review.	true	Boolean, not null
dispensed_amount	Actual cash dispensed. May be less than requested if ATM ran low on cash mid-dispense.	5000.0000	DECIMAL(20,4)
reject_reason	Null on success. Reason code if ATM declined.	null	ENUM: INSUFFICIENT_CASH, CARD_BLOCKED, DAILY_LIMIT_EXCEEDED, null

Entity: ChequeTransaction

Details for cheque-based transactions. Cheques enter a clearing workflow (CTS — Cheque Truncation System). MICR code identifies the originating branch.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
cheque_txn_id	UUID PK.	cheque_txn_uuid	UUID, PK
txn_id	FK to parent Transaction.	txn_uuid	UUID, FK → Transaction
cheque_number	6-digit cheque number from the cheque book.	001234	String(6), not null
drawee_bank_ifsc	IFSC of the bank on which the cheque was drawn.	HDFC0001234	String(11), not null
micr_code	9-digit MICR code encoding bank, city, and branch.	400240006	String(9), not null

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
clearing_date	Date when the cheque is expected to clear via CTS.	2026-05-11	Date, not null
status	Cheque clearing status.	PRESENTED	ENUM: PRESENTED, IN_CLEARING, CLEARED, BOUNCED, RETURNED

Entity: StandingInstruction

A scheduled automatic transaction executed periodically without customer action. Used for recurring rent, EMI, SIP, utility bills. Executed by nightly SI batch job.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
si_id	UUID PK.	si_uuid	UUID, PK
account_id	FK to source Account from which money is debited.	acc_uuid	UUID, FK → Account
beneficiary_id	FK to AccountBeneficiary who receives the transfer.	ben_uuid	UUID, FK → AccountBeneficiary
amount	Fixed amount to transfer each period.	25000.0000	DECIMAL(20,4), check > 0
frequency	How often the SI executes.	MONTHLY	ENUM: DAILY, WEEKLY, MONTHLY, QUARTERLY, YEARLY
next_run_date	Date of next scheduled execution. Updated after each run.	2026-06-01	Date, not null, indexed
max_executions	Maximum number of times this SI should run. Null = unlimited.	12	Integer, nullable
executed_count	How many times SI has been successfully executed so far.	3	Integer, default: 0
status	SI lifecycle state. FAILED if last execution failed (insufficient balance).	ACTIVE	ENUM: ACTIVE, PAUSED, CANCELLED, COMPLETED, FAILED

Entity: TransactionLimit

Per-account, per-channel transaction limits. Customisable by customer within bank-defined maximums. Reset daily at midnight.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
limit_id	UUID PK.	limit_uuid	UUID, PK
account_id	FK to Account.	acc_uuid	UUID, FK → Account, indexed
channel	Which channel these limits apply to.	INTERNET_BANKING	ENUM: ATM, INTERNET_BANKING, MOBILE_APP, UPI, NEFT, RTGS, IMPS, POS
daily_limit	Maximum total amount transactable via this channel in a calendar day.	100000.0000	DECIMAL(20,4)
per_txn_limit	Maximum single transaction amount via this channel.	50000.0000	DECIMAL(20,4)
monthly_limit	Maximum total amount transactable via this channel in a calendar month.	500000.0000	DECIMAL(20,4)
last_reset	Date when daily counters were last reset (midnight job).	2026-05-09	Date, not null

Entity: IdempotencyRecord

Ensures safe retry of payment APIs. Client sends Idempotency-Key header; if seen within 24 hours, cached response returned without re-processing. Stored in Redis with 24-hour TTL.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
key	Client-supplied idempotency key from request header. UUID format recommended. Redis primary key.	uuid-from-client	String(36), Redis key, TTL 24h
txn_id	Transaction ID of the already-processed request. Returned on duplicate requests.	txn_uuid	UUID, Redis value field
created_at	When this idempotency record was first stored.	2026-05-09T10:30:00Z	Timestamp, Redis field
expires_at	Expiry time for this record.	2026-05-10T10:30:00Z	Timestamp, Redis TTL

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/transactions	Initiate a transaction (idempotent — requires Idempotency-Key header)
GET	/api/v1/transactions/{txnId}	Get transaction status and details
GET	/api/v1/transactions?accountId=&from;=&to;=	Transaction history with date range filter
POST	/api/v1/transactions/{txnId}/reverse	Initiate reversal of a posted transaction
GET	/api/v1/transactions/reference/{refNumber}	Look up by bank reference number
POST	/api/v1/transactions/cash/deposit	Branch teller cash deposit
POST	/api/v1/transactions/cash/withdrawal	Branch teller cash withdrawal
POST	/api/v1/transactions/atm	ATM transaction processing (internal, from ATM Switch)
POST	/api/v1/transactions/cheque/deposit	Cheque deposit — enters clearing queue
GET	/api/v1/transactions/cheque/{chequeId}/status	Cheque clearing status
GET	/api/v1/accounts/{accountId}/standing-instructions	List standing instructions
POST	/api/v1/accounts/{accountId}/standing-instructions	Create standing instruction
PUT	/api/v1/standing-instructions/{siId}	Modify SI amount/frequency
DELETE	/api/v1/standing-instructions/{siId}	Cancel standing instruction
GET	/api/v1/accounts/{accountId}/limits	Get transaction limits per channel
PUT	/api/v1/accounts/{accountId}/limits	Update transaction limits
GET	/api/v1/ledger/gl-accounts	General ledger account summary (finance team)
GET	/api/v1/ledger/journal?date=	Journal entries for a given date

Kafka Topics

Direction	Topic Name	Trigger / Consumer
PUBLISHES	transaction.initiated	Fraud Detection — real-time scoring before posting
PUBLISHES	transaction.completed	Account Svc (balance update), Notification, Audit, Reporting
PUBLISHES	transaction.failed	Notification Service — alert customer
PUBLISHES	transaction.reversed	Account Service + Audit Service
PUBLISHES	standing.instruction.executed	Notification Service
SUBSCRIBES	fraud.transaction.blocked	Mark transaction as BLOCKED, trigger reversal
SUBSCRIBES	account.fd.matured	Auto-credit FD maturity proceeds

Key Design Decisions

- Two-phase commit avoided: Account debit → publish event → Account credit (Saga choreography)
- Idempotency-Key header mapped to Redis key; duplicate requests return cached response within 24 hours
- Java 21 virtual threads (Project Loom) handle 50,000 concurrent I/O-bound transaction threads
- PostgreSQL partitioned by transaction_date (monthly partitions) for hot data; Cassandra for cold (>90 days)
- Double-entry: every POST to /transactions creates exactly 2 JournalEntry rows — enforced by DB constraint
- ATM Switch connects via ISO 8583 protocol adapter that converts to REST for this service

06 Payments & Fund Transfer Service

UPI · NEFT · RTGS · IMPS · NACH · Bharat Bill Pay · SWIFT

Purpose & Responsibility

Handles all inter-bank and intra-bank payment rails: UPI (via NPCI UPI switch), NEFT (batch settlement via RBI NEFT), RTGS (real-time gross settlement for amounts $\geq$ INR 2 lakh), IMPS (24x7 instant transfer), NACH (National Automated Clearing House for bulk mandates — EMI, salary, utility), Bharat Bill Payment System (BBPS) for bill payments, and international SWIFT transfers. Each payment rail has its own settlement cycle, failure handling, and compliance requirements.

Architectural Rationale

Payment rails are externally regulated systems with distinct protocols, SLAs, and settlement windows. A single service coordinates the routing logic: selecting the appropriate rail based on amount, time, and beneficiary bank. Separating this from transaction processing allows us to integrate new payment rails (e.g. CBDC in future) without touching the core ledger. The service maintains a Saga with compensation for every external payment to handle partial failures.

Domain Entities

Entity: PaymentOrder

Root entity for any outbound payment regardless of rail. Status tracks payment through its lifecycle. Saga compensation triggered on failures.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
payment_id	UUID PK. Returned to client immediately for status polling.	pmt_uuid	UUID, PK, indexed
customer_id	Who initiated the payment. Used for fraud scoring and limit checks.	cust_uuid	UUID, FK → Customer
debit_account	11-digit account number being debited. Validated for ACTIVE status and sufficient balance.	14000000017	String(11), FK → Account.account_number
beneficiary_id	FK to AccountBeneficiary. The recipient of the payment.	ben_uuid	UUID, FK → AccountBeneficiary
amount	Payment amount. Rail-specific min/max enforced (RTGS: min INR 2L; IMPS: max INR 5L).	50000.0000	DECIMAL(20,4), check > 0
currency	Payment currency. INR for domestic; foreign currency for SWIFT.	INR	String(3), ISO 4217
rail	Payment network selected for routing.	IMPS	ENUM: UPI, NEFT, RTGS, IMPS, NACH, BBPS, SWIFT, INTERNAL
status	Payment state machine.	COMPLETED	ENUM: INITIATED, PENDING_FRAUD_CHECK, PROCESSING, COMPLETED, FAILED, REFUNDED, CANCELLED
npci_ref	NPCI-issued reference for UPI/IMPS transactions.	NPCI2026050900001	String, nullable — only for NPCI rails
rbi_ref	RBI-issued reference for NEFT/RTGS transactions.	RTGS2026050900001	String, nullable — only for RBI rails
initiated_at	Timestamp of payment initiation.	2026-05-09T10:30:00Z	Timestamp, not null
settled_at	Timestamp of final settlement. Null until COMPLETED.	2026-05-09T10:30:05Z	Timestamp, nullable
failure_reason	Populated on FAILED status. Rail-specific error codes.	BENEFICIARY_BANK_OFFLINE	String, nullable

Entity: UpiVpa

Customer's UPI Virtual Payment Address — their unique payment identity on the UPI network (e.g. rohan@nivesh). One customer can have multiple VPAs.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
vpa_id	UUID PK.	vpa_uuid	UUID, PK
customer_id	FK to Customer who owns this VPA.	cust_uuid	UUID, FK → Customer
account_id	FK to Account linked to this VPA. Payments to the VPA credit this account.	acc_uuid	UUID, FK → Account

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
vpa_address	The UPI VPA string. Unique across all banks on UPI network.	rohan@nivesh	String, unique globally via NPCI
is_default	True if this is the customer's primary VPA for receiving payments.	true	Boolean, default: false
linked_at	Timestamp when VPA was linked to the account.	2026-01-20T10:00:00Z	Timestamp, not null
status	VPA status.	ACTIVE	ENUM: ACTIVE, SUSPENDED, DELETED

Entity: UpiTransaction

UPI-specific transaction details linked to a *PaymentOrder*. Stores NPCI-issued transaction IDs and QR code data for P2M transactions.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
upi_txn_id	UUID PK.	upi_txn_uuid	UUID, PK
payment_id	FK to <i>PaymentOrder</i> .	pmt_uuid	UUID, FK → <i>PaymentOrder</i>
payer_vpa	VPA of the payer (our customer).	rohan@nivesh	String, not null
payee_vpa	VPA of the recipient.	priya@hdfc	String, not null
amount	Amount transferred via UPI.	500.0000	DECIMAL(20,4)
txn_id_npci	NPCI-issued 12-digit UPI transaction ID. Used for dispute resolution with NPCI.	123456789012	String(12), unique
status	UPI-specific status from NPCI response.	SUCCESS	ENUM: PENDING, SUCCESS, FAILURE, DEEMED
qr_code	QR code data for P2M transactions. Null for P2P.	upi://pay?pa=merchant@bank...	String, nullable

Entity: NachMandate

Authorises a creditor to debit customer's account automatically on scheduled dates. Registered via NPCI NACH platform. Once active, auto-debits run without further customer action.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
mandate_id	UUID PK.	mand_uuid	UUID, PK
customer_id	Account holder who granted the debit authority.	cust_uuid	UUID, FK → Customer
account_id	Account from which auto-debits will be made.	acc_uuid	UUID, FK → Account
creditor_name	Name of entity authorised to debit (lender, AMC, insurer).	HDFC Home Loans	String, not null
amount	Maximum amount authorised per debit. Actual debit cannot exceed this.	25000.0000	DECIMAL(20,4), max_debit_amount
frequency	How often debits are scheduled.	MONTHLY	ENUM: DAILY, WEEKLY, MONTHLY, QUARTERLY, YEARLY, AS_AND_WHEN
start_date	Date from which auto-debits can begin.	2026-02-01	Date, not null
end_date	Date after which no further debits allowed. Null = no end date.	2036-02-01	Date, nullable
umrn	Unique Mandate Reference Number assigned by NPCI.	NACH000000042360	String(20), unique, assigned by NPCI
status	Mandate lifecycle state.	ACTIVE	ENUM: PENDING_REGISTRATION, ACTIVE, PAUSED, CANCELLED, EXPIRED
bank_approval_status	Whether Nivesh Bank approved this mandate.	APPROVED	ENUM: PENDING, APPROVED, REJECTED

Entity: NachTransaction

Individual auto-debit execution record under a *NACH* mandate. Each scheduled debit creates one record. Failed debits are retried per NPCI guidelines.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
nach_txn_id	UUID PK.	nach_txn_uuid	UUID, PK
mandate_id	FK to <i>NachMandate</i> .	mand_uuid	UUID, FK → <i>NachMandate</i>

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
amount	Actual amount debited (may be less than mandate maximum).	25000.0000	DECIMAL(20,4)
debit_date	Date on which the debit was scheduled and executed.	2026-05-01	Date, not null
status	Execution status of this debit.	SUCCESS	ENUM: SCHEDULED, SUCCESS, FAILED, RETURNED
failure_reason	Null on success. Reason for failure.	null	String, nullable
retry_count	Number of times this debit has been retried after failure.	0	Integer, default: 0

Entity: BillPayment

Tracks a bill payment made via BBPS. Covers utility bills, insurance premiums, loan EMIs, school fees, and more.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
bill_id	UUID PK.	bill_uuid	UUID, PK
customer_id	FK to Customer who paid the bill.	cust_uuid	UUID, FK → Customer
biller_id	BBPS biller identifier. Registered biller on BBPS platform.	MAHADISCO001	String, not null
consumer_number	Customer's account number with the biller (e.g. electricity account number).	1234567890	String, not null
amount	Amount paid for the bill.	1500.0000	DECIMAL(20,4)
due_date	Bill's original due date.	2026-05-15	Date, nullable
paid_at	Timestamp of successful payment.	2026-05-09T11:00:00Z	Timestamp, nullable
bbps_txn_id	BBPS-issued transaction ID. Used for dispute resolution.	BBPS2026050900001	String, unique
status	Payment status.	SUCCESS	ENUM: PENDING, SUCCESS, FAILED, REFUNDED

Entity: SwiftMessage

Details of an international SWIFT transfer. MT103 is the standard format for customer credit transfers. Message built, signed by HSM, and sent via SWIFT Alliance Gateway.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
swift_id	UUID PK.	swift_uuid	UUID, PK
payment_id	FK to PaymentOrder.	pmt_uuid	UUID, FK → PaymentOrder
message_type	SWIFT message type. MT103 = standard customer credit transfer.	MT103	ENUM: MT103, MT202, MT900, MT910
sender_bic	Bank Identifier Code of Nivesh Bank.	NVSHININ	String(11), BIC format
receiver_bic	BIC of the beneficiary bank.	HSBCGB2L	String(11)
value_date	Date on which funds should be credited to the beneficiary.	2026-05-11	Date, not null
charges_type	Who bears SWIFT transfer charges.	SHA	ENUM: OUR (sender pays all), BEN (beneficiary pays), SHA (shared)
status	Message delivery status.	ACKNOWLEDGED	ENUM: DRAFTED, SENT, ACKNOWLEDGED, SETTLED, RETURNED

Entity: PaymentSchedule

Future-dated or recurring payment instruction. Customer can schedule a payment for a specific date or recurring frequency. Executed by the payment scheduler batch job.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
schedule_id	UUID PK.	sched_uuid	UUID, PK
customer_id	FK to Customer who created the schedule.	cust_uuid	UUID, FK → Customer
amount	Amount to transfer on each scheduled execution.	10000.0000	DECIMAL(20,4)
beneficiary_id	FK to AccountBeneficiary.	ben_uuid	UUID, FK → AccountBeneficiary

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
frequency	For recurring payments. Null for one-time future-dated payments.	MONTHLY	ENUM: ONE_TIME, DAILY, WEEKLY, MONTHLY, YEARLY, nullable
next_run_date	Date of next scheduled execution.	2026-06-01	Date, not null
rail	Which payment rail to use.	NEFT	ENUM: UPI, NEFT, RTGS, IMPS, INTERNAL
description	Narration for the scheduled payment.	Monthly rent to landlord	String, nullable

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/payments/upi/send	Initiate UPI payment (P2P or P2M)
POST	/api/v1/payments/upi/collect	Raise UPI collect request on payer's VPA
POST	/api/v1/payments/upi/vpa/register	Register/link a VPA to an account
GET	/api/v1/payments/upi/vpa/resolve?vpa=	Resolve VPA to beneficiary name (before sending)
POST	/api/v1/payments/upi/vpa/link-account	Link bank account to existing VPA
POST	/api/v1/payments/neft	Initiate NEFT transfer (batch, RBI settlement windows)
POST	/api/v1/payments/rtgs	Initiate RTGS transfer (min INR 2L, real-time)
POST	/api/v1/payments/imps	Initiate IMPS (24x7, up to INR 5L)
GET	/api/v1/payments/{paymentId}/status	Poll payment status (PENDING/COMPLETED/FAILED)
GET	/api/v1/payments?accountId=&rail;=	List payment history by account and rail
POST	/api/v1/payments/nach/mandate	Register NACH mandate (EMI/SIP auto-debit)
GET	/api/v1/payments/nach/mandates?customerId=	List all NACH mandates for customer
PUT	/api/v1/payments/nach/mandates/{id}/cancel	Cancel a NACH mandate
GET	/api/v1/payments/bills/billers	List BBPS billers by category
POST	/api/v1/payments/bills/fetch	Fetch bill details from BBPS biller
POST	/api/v1/payments/bills/pay	Pay bill via BBPS
POST	/api/v1/payments/swift	Initiate international SWIFT remittance
GET	/api/v1/payments/exchange-rates	Current forex rates for SWIFT transfers
POST	/api/v1/payments/schedule	Schedule a future or recurring payment
GET	/api/v1/payments/schedule?customerId=	List scheduled payments
DELETE	/api/v1/payments/schedule/{id}	Cancel a scheduled payment

Kafka Topics

Direction	Topic Name	Trigger / Consumer
PUBLISHES	payment.initiated	Fraud Detection — pre-payment risk scoring
PUBLISHES	payment.completed	Transaction Service, Notification, Audit
PUBLISHES	payment.failed	Notification Service — with failure reason
PUBLISHES	nach.debit.processed	Transaction Service — record SI debit
SUBSCRIBES	fraud.payment.blocked	Cancel payment, trigger refund

Key Design Decisions

- UPI integration: bank as PSP — connects to NPCI UPI switch via ISO 20022 / UPI spec APIs
- NEFT batches run every 30 min (8 AM–7 PM weekdays); service queues orders and submits batch to RBI
- RTGS uses RBI's SFMS (Structured Financial Messaging System) adapter for real-time settlement
- Saga compensation: if RTGS debit succeeds but credit fails, automated compensation debit-reversal initiated
- Forex rate sourced from Reuters/Bloomberg API, cached in Redis, stale-while-revalidate strategy
- SWIFT MT103 messages built, signed (HSM), and sent via SWIFT Alliance Gateway integration

07 Loan & Credit Management Service

Home Loan · Personal · Auto · Education · Overdraft · NPA

Purpose & Responsibility

End-to-end loan lifecycle management: application intake, credit score fetch (CIBIL/Experian), automated underwriting with rule-engine scoring, sanction, legal/technical verification for home loans, loan disbursal, EMI scheduling (reducing balance method), prepayment, foreclosure, NPA classification (per RBI IRACP norms), and recovery management. Products: Home Loan, Personal Loan, Auto Loan, Education Loan, Loan Against Property, Overdraft, and Gold Loan.

Architectural Rationale

Lending is the primary revenue driver for any bank. A dedicated service allows: sophisticated credit decisioning models (ML-based scoring in future), independent regulatory reporting (RBI's CRILC, Section 42 CIC returns), strict audit trails for sanction committee decisions, and fine-grained NPA provisioning calculations without impacting core banking speed.

Domain Entities

Entity: LoanApplication

Tracks a loan application from submission through underwriting to final decision. Workflow: customer submission → document upload → credit assessment → RM/committee decision.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
app_id	UUID PK. Reference given to customer for tracking application status.	app_uuid	UUID, PK
customer_id	FK to Customer who applied.	cust_uuid	UUID, FK → Customer
product_type	Type of loan applied for. Determines underwriting rules and documentation requirements.	HOME_LOAN	ENUM: HOME_LOAN, PERSONAL_LOAN, AUTO_LOAN, EDUCATION_LOAN, LAP, OVERDRAFT, GOLD_LOAN
requested_amount	Amount requested. May differ from sanctioned amount based on eligibility.	5000000.0000	DECIMAL(20,4), not null
tenure_months	Requested loan tenure in months.	240	Integer, not null
purpose	Stated purpose of the loan.	Purchase of residential property	String, nullable
co_applicant_id	Optional co-applicant customer_id. Increases combined income eligibility.	cust_uuid_2	UUID, nullable FK → Customer
collateral_type	Type of security offered. Null for unsecured loans.	PROPERTY	ENUM: PROPERTY, VEHICLE, GOLD, SECURITIES, FD, null
status	Application state machine.	UNDER_REVIEW	ENUM: SUBMITTED, DOCUMENT_PENDING, UNDER_REVIEW, CREDIT_ASSESSED, SANCTIONED, REJECTED, DISBURSED, WITHDRAWN
submitted_at	Timestamp of application submission.	2026-05-01T10:00:00Z	Timestamp, not null
decision_at	Timestamp of final sanction/rejection decision.	2026-05-05T15:00:00Z	Timestamp, nullable
decision_by	Who made the final decision. AUTO for automated; emp_id for RM/Credit Committee.	emp_00456	String — AUTO or employee_id

Entity: CreditAssessment

Result of credit bureau checks and underwriting calculations. Both CIBIL and Experian scores fetched; average used for decision.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
assessment_id	UUID PK.	assess_uuid	UUID, PK
app_id	FK to LoanApplication.	app_uuid	UUID, FK → LoanApplication
cibil_score	Credit score from CIBIL. Range 300–900. Below 650 typically rejected.	750	Integer(300-900)
experian_score	Credit score from Experian. Range 300–850.	720	Integer(300-850)

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
dti_ratio	Debt-to-Income ratio: (total EMIs / monthly income) * 100. Max 50% per policy.	35.50	DECIMAL(5,2), percent
foir	Fixed Obligation to Income Ratio. Includes rent, insurance premiums, etc.	40.00	DECIMAL(5,2), percent
loan_to_value	Loan amount as percentage of collateral value. Max 80% for home loans.	75.00	DECIMAL(5,2), percent
risk_grade	Internal risk grade assigned by underwriting engine.	A	ENUM: A (lowest risk), B, C, D, E (highest risk)
recommendation	Underwriting engine's recommendation.	APPROVE	ENUM: APPROVE, REJECT, REFER_TO_COMMITTEE
assessed_at	Timestamp of assessment.	2026-05-04T10:00:00Z	Timestamp, not null

Entity: LoanAccount

Created when a loan is disbursed. Tracks live loan position: outstanding principal, next EMI, DPD counter, and NPA classification. Updated every time an EMI is paid.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
loan_account_id	UUID PK. Customer-facing loan reference number.	loan_uuid	UUID, PK
app_id	FK to LoanApplication from which this account was created.	app_uuid	UUID, FK → LoanApplication
customer_id	FK to primary borrower.	cust_uuid	UUID, FK → Customer
product_type	Type of loan (copied from application at disbursal).	HOME_LOAN	ENUM: same as LoanApplication.product_type
principal	Original disbursed amount. Immutable after disbursal.	4500000.0000	DECIMAL(20,4), immutable
interest_rate	Annual interest rate at origination. For floating loans, updated when RBI repo rate changes.	8.75	DECIMAL(5,2)
tenure_months	Original loan tenure in months.	240	Integer, not null
emi_amount	Monthly EMI calculated at origination. Recalculated on prepayment.	44745.0000	DECIMAL(20,4)
outstanding_principal	Remaining principal. Decreases with each EMI. Used for foreclosure quotes.	4320000.0000	DECIMAL(20,4)
disbursed_at	Timestamp of loan disbursal.	2026-05-10T10:00:00Z	Timestamp, not null
maturity_date	Date of final EMI payment.	2046-05-10	Date, not null
status	Loan account state.	ACTIVE	ENUM: ACTIVE, CLOSED, NPA, WRITTEN_OFF, LEGAL_ACTION
npa_date	Date loan was classified as NPA (90+ DPD). Null for performing loans.	null	Date, nullable — non-null only if status=NPA

Entity: EmiSchedule

Pre-generated amortisation schedule showing every EMI across the loan tenure. Updated when prepayment changes outstanding balance. Status tracks collection per installment.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
schedule_id	UUID PK.	sched_uuid	UUID, PK
loan_account_id	FK to LoanAccount.	loan_uuid	UUID, FK → LoanAccount
installment_number	Sequential number from 1 to total tenure months.	1	Integer, not null
due_date	Date by which EMI must be paid.	2026-06-10	Date, not null, indexed
principal_component	Portion of this EMI that reduces principal. Increases with each installment.	11995.0000	DECIMAL(20,4)
interest_component	Portion that pays interest. Decreases with each installment.	32750.0000	DECIMAL(20,4)
total_emi	Total EMI = principal_component + interest_component.	44745.0000	DECIMAL(20,4)
paid_amount	Actual amount paid. May differ from total_emi for partial payments.	44745.0000	DECIMAL(20,4), nullable

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
paid_date	Actual date payment was received. Null if PENDING or OVERDUE.	2026-06-08	Date, nullable
status	Payment status of this installment.	PENDING	ENUM: PENDING, PAID, OVERDUE, PARTIALLY_PAID, WAIVED

Entity: Collateral

Security offered against the loan. For home loans: property documents. For gold loans: gold valuation. Collateral insurance tracked here.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
collateral_id	UUID PK.	collateral_uuid	UUID, PK
loan_account_id	FK to LoanAccount.	loan_uuid	UUID, FK → LoanAccount
type	Type of collateral.	PROPERTY	ENUM: PROPERTY, VEHICLE, GOLD, SECURITIES, FD
description	Detailed description of the collateral asset.	2BHK Flat, Orchid Heights, Pune	String, not null
value	Appraised market value of the collateral.	6000000.0000	DECIMAL(20,4)
valuation_date	Date of last appraisal. Re-valuation required if > 3 years old.	2026-04-15	Date, not null
valuation_by	Name of the approved valuer/appraisal agency.	Knight Frank India	String, not null
insurance_expiry	Expiry date of collateral insurance policy. System alerts 60 days before expiry.	2027-01-01	Date, nullable

Entity: Disbursement

Records actual disbursal of loan funds to customer. Partial disbursements possible for construction-linked home loans (disbursed in stages as construction progresses).

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
disbursement_id	UUID PK.	disb_uuid	UUID, PK
loan_account_id	FK to LoanAccount.	loan_uuid	UUID, FK → LoanAccount
amount	Amount disbursed in this tranche.	4500000.0000	DECIMAL(20,4)
disbursement_date	Date funds were credited to customer's account.	2026-05-10	Date, not null
credit_account_id	Account ID where funds were credited.	acc_uuid	UUID, FK → Account
mode	How the funds were transferred.	RTGS	ENUM: RTGS, CHEQUE, ACCOUNT_CREDIT, DD
reference	Payment reference number from the transaction.	RTGS2026051000001	String, not null

Entity: Prepayment

Records partial prepayment of loan principal. Customer can reduce EMI (same tenure) or reduce tenure (same EMI). Penalty applies for fixed-rate loans within lock-in period.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
prepayment_id	UUID PK.	prepay_uuid	UUID, PK
loan_account_id	FK to LoanAccount.	loan_uuid	UUID, FK → LoanAccount
amount	Prepayment amount applied to reduce outstanding principal after penalty deduction.	500000.0000	DECIMAL(20,4)
prepayment_date	Date prepayment was made.	2026-05-09	Date, not null
penalty_amount	Penalty charged. 0% for floating-rate (RBI mandate); 2% for fixed-rate within lock-in.	0.0000	DECIMAL(20,4), default: 0
updated_outstanding	New outstanding principal after this prepayment.	3820000.0000	DECIMAL(20,4)
updated_tenure	New loan tenure after prepayment (if customer chooses to reduce tenure).	200	Integer, nullable — null if EMI reduced instead

Entity: NpaRecord

Created when loan crosses 90 DPD and classified as NPA per RBI IRACP norms. Provisioning percentages determined by sub-classification.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
npa_id	UUID PK.	npa_uuid	UUID, PK
loan_account_id	FK to LoanAccount.	loan_uuid	UUID, FK → LoanAccount
classification	NPA sub-category per RBI IRACP norms. Determines mandatory provisioning percentage.	SUBSTANDARD	ENUM: SUBSTANDARD (0-12m, 15% provision), DOUBTFUL_I (12-24m, 25%), DOUBTFUL_II (24-36m, 40%), DOUBTFUL_III (>36m, 100%), LOSS
classified_at	Date NPA classification was applied.	2026-03-01	Date, not null
provision_amount	Rupee amount set aside by the bank as provision. Required by RBI.	675000.0000	DECIMAL(20,4)
recovery_agent_id	Employee or external agency assigned for recovery.	agent_emp_id	String, nullable
last_contact_at	Last date recovery agent contacted the customer.	2026-04-15	Date, nullable

Entity: LegalVerification

Legal due diligence record for property collateral. Advocate examines title documents and gives a clear title opinion. Required before disbursal of home loans.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
legal_id	UUID PK.	legal_uuid	UUID, PK
collateral_id	FK to Collateral being verified.	collateral_uuid	UUID, FK → Collateral
advocate_name	Name of the empanelled legal advocate who conducted the title search.	Adv. Suresh Patil	String, not null
opinion_date	Date when legal opinion was issued.	2026-04-20	Date, not null
clear_title	True if title is clear of encumbrances. False blocks disbursal.	true	Boolean, not null
encumbrance_status	Any existing encumbrances, mortgages, or disputes on the property.	CLEAR	ENUM: CLEAR, ENCUMBERED, DISPUTED, PENDING_VERIFICATION
remarks	Advocate's detailed remarks about the title search findings.	Clear title, no encumbrances for past 30 years.	String(1000), nullable

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/loans/applications	Submit new loan application
GET	/api/v1/loans/applications/{appId}	Get application status and details
GET	/api/v1/loans/applications?customerId=	List all applications for a customer
POST	/api/v1/loans/applications/{appId}/documents	Upload supporting documents (income proof, property docs)
GET	/api/v1/loans/applications/{appId}/credit-report	Fetch CIBIL/Experian score for application
PUT	/api/v1/loans/applications/{appId}/decision	Sanction/reject application (RM/Credit Officer)
POST	/api/v1/loans/applications/{appId}/disburse	Trigger loan disbursement to customer account
GET	/api/v1/loans/{loanId}	Get loan account details
GET	/api/v1/loans/{loanId}/emi-schedule	Full EMI amortisation schedule
GET	/api/v1/loans/{loanId}/outstanding	Current outstanding with interest to date
POST	/api/v1/loans/{loanId}/prepayment	Partial prepayment — recalculates EMI or tenure
POST	/api/v1/loans/{loanId}/foreclosure	Full foreclosure with penalty calculation
GET	/api/v1/loans/{loanId}/foreclosure-quote	Get foreclosure amount valid for 3 days
GET	/api/v1/loans/{loanId}/statement	Loan account statement — all debits and credits
POST	/api/v1/loans/{loanId}/noc	Generate No Objection Certificate after closure
GET	/api/v1/loans/npa?branch=&classification;=	NPA report for branch/region (compliance)
POST	/api/v1/loans/{loanId}/emi/waiver	Late fee/penalty waiver request

Method	Endpoint	Purpose
GET	/api/v1/loans/products	List available loan products with current rates
GET	/api/v1/loans/emi-calculator?amount=&rate;=&tenure;=	EMI calculator (public endpoint, no auth)

Kafka Topics

Direction	Topic Name	Trigger / Consumer
PUBLISHES	loan.disbursed	Transaction Service — credit disbursed amount to customer account
PUBLISHES	loan.emi.overdue	Notification Service + Fraud Detection
PUBLISHES	loan.npa.classified	Reporting + Compliance Service
PUBLISHES	loan.closed	Notification + Audit Service
SUBSCRIBES	transaction.completed	Mark EMI as PAID when payment received

Key Design Decisions

- Credit score fetched from CIBIL API + Experian API; average used; both stored per assessment
- EMI calculated using standard reducing-balance formula — pure Java math, no float arithmetic
- NPA classification batch job runs nightly: 90+ DPD → Substandard; 12 months → Doubtful
- Prepayment penalty: 0% for floating-rate loans (RBI mandate); 2% for fixed-rate within lock-in
- Sanction authority: up to INR 5L auto-approved; INR 5L-50L RM approves; INR 50L+ Credit Committee
- CRILC (Central Repository of Information on Large Credits) reporting automated via scheduled batch

08 Card Management Service

Debit · Credit · Prepaid · Virtual cards · Tokenization · PCI-DSS L1

Purpose & Responsibility

Full lifecycle management of payment cards: Debit cards (linked to savings/current accounts), Credit cards (revolving credit with billing cycle, minimum due, credit limit), Prepaid cards (gift/travel cards), and Virtual cards (for online transactions). Covers card issuance (physical + virtual), PIN management (HSM-encrypted), card controls (domestic/international/online/contactless toggle), EMI-on-card conversion, reward points, card-on-file tokenization (PCI-DSS compliant), lost/stolen card blocking, and chargeback management.

Architectural Rationale

Card operations are PCI-DSS Level 1 compliant — the strictest payment security standard. A dedicated service maintains PCI scope boundary: cardholder data (PAN, CVV, expiry) never leaves this service unencrypted. Integration with Visa/Mastercard/RuPay networks requires certified HSM for PIN translation and cryptogram generation.

Domain Entities

Entity: Card

Master card record. Full PAN never stored — only last 4 digits and a network token. Actual PAN exists only within the HSM. This design maintains PCI-DSS scope boundary.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
card_id	UUID PK. Internal reference for card operations.	card_uuid	UUID, PK
customer_id	FK to Customer who owns this card.	cust_uuid	UUID, FK → Customer
account_id	FK to Account linked to this card. Debit cards link to savings; credit cards to CreditCardAccount.	acc_uuid	UUID, FK → Account
card_number_tokenized	EMVCo-compliant token from Visa/MC/RuPay Token Service. Not the real PAN. Safe to store.	TOKEN_VALUE_FROM_TSP	String, from Token Service Provider
last4	Last 4 digits of actual PAN. Used for display purposes only. Safe to store.	4521	String(4), not null
card_type	Type of card.	DEBIT	ENUM: DEBIT, CREDIT, PREPAID, VIRTUAL
network	Payment network. Determines interchange routing and tokenization provider.	RUPAY	ENUM: VISA, MASTERCARD, RUPAY, AMEX
status	Card lifecycle state. HOTLISTED is permanent block — cannot be unblocked; new card issued.	ACTIVE	ENUM: ACTIVE, BLOCKED, HOTLISTED, EXPIRED, REPLACED, PENDING_DISPATCH
expiry_month	Card expiry month displayed on physical card.	05	Integer(1-12), not null
expiry_year	Card expiry year.	2029	Integer, not null
emboss_name	Name printed on physical card. Truncated to 26 chars.	ROHAN SHARMA	String(26), not null
issued_at	Timestamp of card issuance.	2026-01-20T10:00:00Z	Timestamp, not null
physical_dispatched	True once physical card dispatched via courier. Virtual cards always false.	true	Boolean, default: false

Entity: CreditCardAccount

Financial account linked to a credit card. Tracks revolving credit position. Billing cycle generates statement on billing_day each month. Payment due 20 days after statement date.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
cc_account_id	UUID PK.	cc_uuid	UUID, PK
card_id	FK to Card.	card_uuid	UUID, FK → Card, unique — one credit account per card
credit_limit	Maximum credit extended. Set based on income and CIBIL score.	200000.0000	DECIMAL(20,4)
available_credit	How much more can be spent: credit_limit - outstanding_balance.	150000.0000	DECIMAL(20,4), derived

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
outstanding_balance	Total amount owed including unbilled transactions.	50000.0000	DECIMAL(20,4)
minimum_due	Minimum payment required. Higher of INR 100 or 5% of outstanding.	2500.0000	DECIMAL(20,4), calculated at statement time
billing_day	Day of month statement is generated. Customer can choose at application time (1-28).	15	Integer(1-28), not null
payment_due_date	Date by which minimum payment must be made. billing_day + 20 days.	2026-06-04	Date, calculated field
interest_rate_monthly	Monthly interest rate on revolving balance. Typically 3.5% = 42% per annum.	3.50	DECIMAL(4,2), percent per month
grace_period_days	Days after billing date before interest starts accruing.	20	Integer, default: 20

Entity: CardTransaction

Records each card-based transaction. Created on authorization and updated when settlement occurs. Feeds into billing cycle aggregation.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
card_txn_id	UUID PK.	card_txn_uuid	UUID, PK
card_id	FK to Card used.	card_uuid	UUID, FK → Card, indexed
merchant_name	Name of the merchant where transaction occurred.	Swiggy India Pvt Ltd	String, not null
merchant_category_code	4-digit ISO 18245 MCC. Used for reward points categorisation.	5812	String(4), MCC code
amount	Transaction amount.	500.0000	DECIMAL(20,4), not null
currency	Transaction currency.	INR	String(3), ISO 4217
txn_type	Type of card transaction.	PURCHASE	ENUM: PURCHASE, REFUND, CASH_ADVANCE, EMI, REVERSAL
auth_code	Authorization code from card network. Used for dispute resolution.	AUTH123456	String(6), not null
status	Transaction status in authorization-settlement lifecycle.	SETTLED	ENUM: AUTHORIZED, PENDING_SETTLEMENT, SETTLED, REVERSED, DISPUTED
txn_at	Timestamp of transaction at merchant.	2026-05-09T12:30:00Z	Timestamp, not null, indexed

Entity: CardControl

Granular on/off switches and limits for a card. Customer can toggle channels instantly via mobile app. Changes take effect in real-time via cache invalidation.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
control_id	UUID PK.	control_uuid	UUID, PK
card_id	FK to Card. One control record per card.	card_uuid	UUID, FK → Card, unique
domestic_enabled	Whether card can be used at domestic POS/ATM. Default: true.	true	Boolean, default: true
international_enabled	Whether card works at international merchants. Default: false per RBI mandate.	false	Boolean, default: false — must explicitly enable
online_enabled	Whether card can be used for e-commerce (Card Not Present) transactions.	true	Boolean, default: true
contactless_enabled	Whether NFC tap-to-pay enabled. Transactions under INR 5,000 do not require PIN.	true	Boolean, default: true
atm_enabled	Whether cash withdrawal at ATMs is enabled.	true	Boolean, default: true
daily_limit	Maximum total spend across all channels per calendar day.	100000.0000	DECIMAL(20,4), customer-configurable within bank max
per_txn_limit	Maximum single transaction amount.	50000.0000	DECIMAL(20,4)
updated_at	Timestamp of last control change. Used for audit trail.	2026-05-09T10:00:00Z	Timestamp, not null

Entity: RewardPoint

Tracks reward points earned and redeemed for a card. Points earned based on spend category and card tier. Expire after 2 years.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
reward_id	UUID PK.	reward_uuid	UUID, PK
card_id	FK to Card.	card_uuid	UUID, FK → Card, unique — one record per card
points_earned	Total points earned since card issuance.	15000	Integer, not null
points_redeemed	Total points redeemed so far.	5000	Integer, default: 0
points_balance	Available points: points_earned - points_redeemed - expired.	10000	Integer, derived
expiry_date	Oldest unredeemed points expire on this date.	2028-01-20	Date, rolling 2-year window
earn_rate_per_100	Points earned per INR 100 spent. Varies by card type.	2	Integer, e.g. 2 points per INR 100
last_updated	Timestamp of last points update.	2026-05-09T10:00:00Z	Timestamp, not null

Entity: CardToken

EMVCo token issued for a specific card-device combination (Apple Pay, Google Pay). Allows contactless payments without exposing real PAN.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
token_id	UUID PK.	token_uuid	UUID, PK
card_id	FK to Card for which this token was issued.	card_uuid	UUID, FK → Card
token_value	Tokenised PAN value issued by the Token Service Provider (Visa/MC/RuPay).	TOKEN_VALUE_FROM_TSP	String, from TSP
token_requestor	Which wallet/platform requested tokenization.	GOOGLE_PAY	ENUM: APPLE_PAY, GOOGLE_PAY, AMAZON, SAMSUNG_PAY
device_id	FK to DeviceFingerprint of the device where token is provisioned.	device_uuid	UUID, FK → DeviceFingerprint
status	Token lifecycle state.	ACTIVE	ENUM: ACTIVE, SUSPENDED, DELETED
created_at	Timestamp of token provisioning.	2026-01-25T09:00:00Z	Timestamp, not null

Entity: Chargeback

A customer dispute for an unauthorised or incorrectly processed card transaction. Bank investigates and provides provisional credit during resolution.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
chargeback_id	UUID PK.	cb_uuid	UUID, PK
card_txn_id	FK to CardTransaction being disputed.	card_txn_uuid	UUID, FK → CardTransaction
reason_code	ISO 8583/card network chargeback reason code.	4853	String — network-specific reason code
description	Customer's description of the dispute.	Transaction not authorized by me	String, not null
amount	Amount being disputed.	1500.0000	DECIMAL(20,4)
filed_at	Timestamp when chargeback was filed.	2026-05-09T14:00:00Z	Timestamp, not null
status	Chargeback resolution status.	OPEN	ENUM: OPEN, PROVISIONAL_CREDIT, MERCHANT_RESPONSE, RESOLVED_CUSTOMER, RESOLVED_MERCHANT
resolution_at	Timestamp of final resolution. Null while open.	null	Timestamp, nullable

Entity: CardStatement

Monthly credit card statement generated on billing_day. PDF stored in S3. Contains all transactions in billing period, minimum due, and total due.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
statement_id	UUID PK.	stmt_uuid	UUID, PK
cc_account_id	FK to CreditCardAccount.	cc_uuid	UUID, FK → CreditCardAccount

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
billing_period_start	First date of billing period.	2026-04-15	Date, not null
billing_period_end	Last date of billing period.	2026-05-14	Date, not null
total_spends	Total amount spent during this billing period.	45000.0000	DECIMAL(20,4)
minimum_due	Minimum amount to pay by due date to avoid late fee.	2500.0000	DECIMAL(20,4)
total_due	Full outstanding balance to avoid interest.	50000.0000	DECIMAL(20,4)
generated_at	Timestamp when statement was generated.	2026-05-15T00:05:00Z	Timestamp, not null
s3_path	S3 path to PDF statement. Pre-signed URL generated on demand.	s3://nvsh-statements/card/stmt_uuid.pdf	String, not null

Entity: EmiConversion

Records conversion of a credit card purchase to EMI. Customer pays in equal monthly instalments with interest instead of one lump sum.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
emi_id	UUID PK.	emi_uuid	UUID, PK
card_txn_id	FK to original CardTransaction being converted to EMI.	card_txn_uuid	UUID, FK → CardTransaction
amount	Original transaction amount being converted.	30000.0000	DECIMAL(20,4)
tenure_months	Number of EMI instalments chosen by customer.	6	Integer, not null
emi_amount	Monthly EMI amount.	5250.0000	DECIMAL(20,4)
interest_rate	Annual interest rate for this EMI conversion.	15.00	DECIMAL(5,2), percent per annum
processing_fee	One-time processing fee charged at conversion.	299.0000	DECIMAL(20,4)
status	EMI conversion status.	ACTIVE	ENUM: ACTIVE, FORECLOSED, COMPLETED

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/cards/debit	Request new debit card (physical + virtual)
POST	/api/v1/cards/credit/apply	Credit card application
GET	/api/v1/cards?customerId=	List all cards for a customer
GET	/api/v1/cards/{cardId}	Card details (masked PAN, controls, limits)
PUT	/api/v1/cards/{cardId}/status	Block/Unblock/Hotlist a card
POST	/api/v1/cards/{cardId}/pin/set	Set new PIN (encrypted via HSM session key)
POST	/api/v1/cards/{cardId}/pin/change	Change PIN (validate old PIN + set new)
POST	/api/v1/cards/{cardId}/pin/reset	Reset forgotten PIN via OTP flow
GET	/api/v1/cards/{cardId}/controls	Get card control settings
PUT	/api/v1/cards/{cardId}/controls	Toggle domestic/international/online/contactless
PUT	/api/v1/cards/{cardId}/limits	Modify daily spend / per-transaction limit
GET	/api/v1/cards/{cardId}/transactions?from=&to;=	Card transaction history with date range
POST	/api/v1/cards/{cardId}/virtual	Generate virtual card number for online use
GET	/api/v1/cards/{cardId}/credit-account	Credit card account details (balance, due date)
GET	/api/v1/cards/{cardId}/statements	List credit card statements
GET	/api/v1/cards/{cardId}/statements/{stmtId}	Download statement PDF from S3
POST	/api/v1/cards/{cardId}/emi	Convert a transaction to EMI
GET	/api/v1/cards/{cardId}/rewards	Reward points balance and history
POST	/api/v1/cards/{cardId}/rewards/redeem	Redeem reward points
POST	/api/v1/cards/{cardId}/chargeback	File a chargeback/dispute
GET	/api/v1/cards/{cardId}/chargebacks	Chargeback history and status

Method	Endpoint	Purpose
POST	/api/v1/cards/{cardId}/add-to-wallet	Tokenize card to Google Pay / Apple Pay

Kafka Topics

Direction	Topic Name	Trigger / Consumer
PUBLISHES	card.transaction.authorized	Fraud Detection — real-time auth scoring
PUBLISHES	card.blocked	Notification + Audit Service
PUBLISHES	credit.bill.generated	Notification Service — statement ready alert
PUBLISHES	credit.payment.due	Notification Service — payment reminder

Key Design Decisions

- Card PAN stored only as last-4 digits + token; full PAN exists only in HSM — PCI-DSS compliant
- PIN encrypted using 3DES/AES under HSM Zone Key; only HSM can decrypt for PIN verification
- Virtual card numbers generated with Luhn-valid PAN, short expiry (1 hour / single use configurable)
- Tokenization follows EMVCo token spec; token requestor onboarding via Visa/MC token service
- Credit card billing cycle: transactions aggregated, statement generated on billing day, due in 20 days
- Minimum due: higher of INR 100 or 5% of outstanding; late payment fee charged if missed

09 Notification Service

SMS · Email · Push · In-App · WhatsApp

Purpose & Responsibility

Multi-channel notification delivery engine. Sends transactional alerts (debit/credit SMS, OTP), marketing notifications (new products, offers), regulatory communications (account statements, tax certificates), and in-app notifications. Channels: SMS (via SMS gateway — Exotel/Textlocal), Email (AWS SES), Push Notifications (FCM/APNs), WhatsApp (via WhatsApp Business API), and In-App notification center. Handles template management, delivery tracking, opt-in/opt-out preferences, and delivery retry.

Architectural Rationale

Notifications touch every customer at every transaction — 500 million customers x multiple events = billions of messages daily. Separating this service means: SMS/email providers can be swapped without touching business logic, delivery failures never impact transaction processing, and customer preferences are centrally managed. MongoDB's flexible schema handles varied notification payloads efficiently.

Domain Entities

Entity: Notification

One notification record per delivery attempt. Tracks the lifecycle from QUEUED through provider dispatch to DELIVERED or FAILED. Retained for compliance and customer support.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
notification_id	UUID PK.	notif_uuid	UUID, MongoDB _id
customer_id	FK to Customer this notification is for.	cust_uuid	UUID, indexed
type	Business event type. Determines template selection and message priority.	DEBIT_ALERT	ENUM: DEBIT_ALERT, CREDIT_ALERT, OTP, LOGIN_ALERT, EML_REMINDER, STATEMENT_READY, CARD_BLOCKED, etc.
channel	Delivery channel. OTP always goes via SMS regardless of preferences.	SMS	ENUM: SMS, EMAIL, PUSH, WHATSAPP, IN_APP
template_id	FK to NotificationTemplate. DLT-registered templates for SMS compliance.	tpl_debit_alert_v3	String, FK → NotificationTemplate._id
payload_json	Template variables for this notification (amount, account last4, merchant, etc.).	{"amount":"500","account":"...0017"}	JSON, not null
status	Delivery lifecycle state. FAILED after 4 retry attempts → pushed to DLQ.	DELIVERED	ENUM: QUEUED, SENDING, SENT, DELIVERED, FAILED, DLQ
sent_at	Timestamp when notification was dispatched to provider.	2026-05-09T10:30:01Z	Timestamp, nullable
delivered_at	Timestamp of confirmed delivery (where provider supports delivery receipts).	2026-05-09T10:30:03Z	Timestamp, nullable
failure_reason	Null on success. Provider error code + description on failure.	PROVIDER_TIMEOUT	String, nullable

Entity: NotificationTemplate

DLT-registered message templates. SMS templates must be pre-registered with TRAI's DLT platform before they can be sent. Templates support variable substitution.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
template_id	MongoDB _id. Short descriptive key.	tpl_debit_alert_v3	String, PK
name	Human-readable template name.	Debit Alert v3	String, not null
channel	Which channel this template is for.	SMS	ENUM: SMS, EMAIL, PUSH, WHATSAPP, IN_APP
subject	Email subject line. Null for SMS/Push.	Your account has been debited	String, nullable
body_template	Message body with placeholder variables in {variable_name} format.	Dear {customer_name}, INR {amount} debited from account ending {last4}.	String, not null

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
language	Language of this template.	ENGLISH	ENUM: ENGLISH, HINDI, MARATHI, TAMIL, TELUGU, KANNADA, BENGALI, GUJARATI, PUNJABI, MALAYALAM
version	Template version number. Old versions kept for audit.	3	Integer, not null
variables_json	List of required template variables and their data types.	[{"customer_name","amount","last4"}]	JSON array
active	Only active templates are used for dispatching.	true	Boolean, default: true

Entity: CustomerPreference

Customer's notification opt-in/opt-out settings. OTP and regulatory notices cannot be opted out. Marketing requires explicit opt-in per DPDPA.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
pref_id	MongoDB _id.	pref_uuid	String, PK
customer_id	FK to Customer.	cust_uuid	UUID, unique, indexed
channel_enabled_map	JSON map of which channels are enabled.	{"SMS":true,"EMAIL":true,"WHATS APP":false,"PUSH":true}	JSON, not null
language	Preferred language for notifications.	HINDI	ENUM: ENGLISH, HINDI, MARATHI, TAMIL, TELUGU, KANNADA, BENGALI, GUJARATI, PUNJABI, MALAYALAM
marketing_opt_in	Whether customer consented to marketing communications. DPDPA compliance field.	false	Boolean, default: false — opt-in required
sms_opt_in	Whether customer receives SMS alerts.	true	Boolean, default: true
email_opt_in	Whether customer receives email alerts.	true	Boolean, default: true
whatsapp_opt_in	Whether customer receives WhatsApp messages.	false	Boolean, default: false
updated_at	Timestamp of last preference update.	2026-05-01T09:00:00Z	Timestamp, not null

Entity: DeviceToken

FCM (Android) or APNs (iOS) push notification token for a customer's mobile device. Tokens change on app reinstall. Updated on every app launch if changed.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
token_id	MongoDB _id.	token_uuid	String, PK
customer_id	FK to Customer.	cust_uuid	UUID, indexed
device_id	FK to DeviceFingerprint.	device_uuid	UUID
platform	Mobile platform.	ANDROID	ENUM: ANDROID, IOS
fcm_token	Firebase Cloud Messaging token for Android. Null for iOS.	FCM_TOKEN_STRING	String, nullable
apns_token	Apple Push Notification Service token for iOS. Null for Android.	APNS_TOKEN_HEX	String, nullable
updated_at	Timestamp of last token update.	2026-05-01T10:00:00Z	Timestamp, not null
active	False when customer logs out or uninstalls app.	true	Boolean, default: true

Entity: DeliveryLog

Detailed log of each delivery attempt to a provider. Multiple records per notification if retries occur. Used for provider SLA reporting and debugging.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
log_id	MongoDB _id.	log_uuid	String, PK
notification_id	FK to Notification.	notif_uuid	UUID, indexed
channel	Which channel was attempted.	SMS	ENUM: SMS, EMAIL, PUSH, WHATSAPP
provider	Which provider was used for this attempt.	EXOTEL	ENUM: EXOTEL, TEXTLOCAL, AWS_SES, FCM, APNS, META_WHATSAPP

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
provider_ref	Provider's message ID for tracing and receipts.	EXOTEL_MSG_12345	String, nullable
attempt_number	Which retry attempt this was. 1 = first attempt.	1	Integer, not null
status	Outcome of this delivery attempt.	SUCCESS	ENUM: SUCCESS, FAILED, TIMEOUT
timestamp	Timestamp of this delivery attempt.	2026-05-09T10:30:01Z	Timestamp, not null

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/notifications/send	Send a notification (internal — called by other services)
GET	/api/v1/notifications?customerId=	In-app notification center: list all notifications
PUT	/api/v1/notifications/{notifId}/read	Mark in-app notification as read
DELETE	/api/v1/notifications/{notifId}	Delete an in-app notification
GET	/api/v1/notifications/preferences/{customerId}	Get notification preferences
PUT	/api/v1/notifications/preferences/{customerId}	Update notification preferences (opt-in/out)
POST	/api/v1/notifications/device-tokens	Register/update FCM/APNs device token
DELETE	/api/v1/notifications/device-tokens/{id}	Remove device token on logout
GET	/api/v1/notifications/templates	List all active templates
POST	/api/v1/notifications/templates	Create new notification template
PUT	/api/v1/notifications/templates/{id}	Update template content
POST	/api/v1/notifications/bulk	Bulk notification send (marketing — admin)

Kafka Topics

Direction	Topic Name	Trigger / Consumer
SUBSCRIBES	transaction.completed	Debit/credit SMS and in-app alert
SUBSCRIBES	payment.completed	Payment success notification
SUBSCRIBES	payment.failed	Payment failure alert
SUBSCRIBES	auth.account.locked	Account lock security alert
SUBSCRIBES	loan.emi.overdue	EMI reminder
SUBSCRIBES	card.blocked	Card block confirmation
SUBSCRIBES	credit.payment.due	Credit card due date reminder

Key Design Decisions

- Kafka consumer processes notification events → renders template → dispatches to channel-specific adapter
- SMS gateway failover: primary Exotel → fallback to Textlocal → fallback to direct DLT-registered gateway
- Retry policy: exponential backoff — 1min, 5min, 30min, 2hr; max 4 attempts; then FAILED with DLQ
- WhatsApp messages via Meta Cloud API — only templated messages (approved by WhatsApp) for compliance
- RBI mandates SMS for every debit > INR 0 and every credit > INR 0 — this service guarantees delivery
- DLT template registration enforced — only TRAI-registered templates sent via SMS

10 Fraud Detection Service

Real-time risk scoring · Velocity checks · ML anomaly detection

Purpose & Responsibility

Real-time and near-real-time fraud prevention. Evaluates every transaction and payment using a rule engine (velocity checks, geo-location mismatch, device fingerprint mismatch, unusual amount patterns) and an ML scoring model (XGBoost-based, served via REST from a model server). Blocks suspicious transactions automatically, generates alerts for manual review, maintains fraud case management, and feeds back confirmed fraud outcomes to retrain the ML model.

Architectural Rationale

Fraud is a real-time problem — detection must happen in under 100ms before a transaction is authorised. A dedicated service with Redis-backed velocity counters and pre-loaded rule sets achieves sub-50ms response times. Isolation means: ML model updates don't require banking downtime, fraud rules are configurable without code deployments, and the service can be scaled aggressively during peak fraud periods (festival seasons, 11 PM–2 AM window).

Domain Entities

Entity: FraudRule

Configurable fraud detection rule written in Drools DSL. Rules can be added, updated, or deactivated without code deployment — only a rule file reload is needed.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
rule_id	UUID PK.	rule_uuid	UUID, PK
rule_code	Short unique code used in alerts and logs.	VELOCITY_5_IN_10MIN	String, unique, not null
description	Human-readable description of what this rule detects.	5 transactions in 10 minutes from same customer	String, not null
condition_dsl	Drools DRL condition expression that triggers the rule.	txnCount > 5 && windowMinutes <= 10	String (Drools DSL), not null
action	What happens when rule fires. BLOCK rejects instantly. ALERT creates case for analyst. MONITOR logs without blocking.	BLOCK	ENUM: BLOCK, ALERT, MONITOR
severity	Risk severity. CRITICAL rules auto-block. HIGH creates immediate review alert.	HIGH	ENUM: LOW, MEDIUM, HIGH, CRITICAL
active	Rules can be deactivated without deletion for A/B testing or emergency rollback.	true	Boolean, default: true
created_by	Employee ID of who created this rule.	emp_00567	String, not null

Entity: RiskScore

Output of the ML model evaluation for a transaction or login event. Score 0-1000. Stored in Cassandra for historical analysis and model retraining pipeline.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
score_id	UUID PK.	score_uuid	UUID, Cassandra PK
entity_id	ID of entity being scored (txn_id, payment_id, or user_id for login).	txn_uuid	UUID, not null
entity_type	Type of entity being scored.	TRANSACTION	ENUM: TRANSACTION, PAYMENT, LOGIN
score	Numeric risk score 0 (no risk) to 1000 (highest). Thresholds: <300 allow, 300-600 monitor, >600 block.	750	Integer(0-1000), not null
risk_level	Categorical bucket derived from score.	HIGH	ENUM: LOW (<300), MEDIUM (300-600), HIGH (600-800), CRITICAL (>800)
contributing_factors_json	Features that contributed most to the score. Used for model explainability.	["new_device", "high_amount", "geo_mismatch"]	JSON array, for XAI
scored_at	Timestamp of scoring. Used for time-series analysis in Cassandra.	2026-05-09T10:30:00.050Z	Timestamp, not null

Entity: FraudAlert

Created when a transaction triggers a fraud rule or exceeds ML risk threshold. Assigned to fraud analysts for review. Outcome feeds back into model training.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
alert_id	UUID PK.	alert_uuid	UUID, PK
customer_id	FK to Customer involved in suspicious activity.	cust_uuid	UUID, FK → Customer
txn_id	FK to Transaction or Payment that triggered this alert.	txn_uuid	UUID, nullable
rule_triggered	Which fraud rule triggered this alert.	VELOCITY_5_IN_10MIN	String, FK → FraudRule.rule_code
risk_score	ML model risk score that contributed to this alert.	750	Integer(0-1000)
status	Alert review lifecycle.	OPEN	ENUM: OPEN, REVIEWED, CONFIRMED_FRAUD, FALSE_POSITIVE
assigned_to	Fraud analyst employee ID assigned for review.	emp_00789	String, nullable
created_at	Timestamp of alert creation.	2026-05-09T10:30:00Z	Timestamp, not null

Entity: FraudCase

Groups multiple related FraudAlerts into a single investigation case. A case may span multiple transactions or time periods involving the same customer.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
case_id	UUID PK.	case_uuid	UUID, PK
alert_ids	Array of FraudAlert IDs grouped into this case.	["alert_uuid_1","alert_uuid_2"]	UUID[], not null
customer_id	FK to Customer under investigation.	cust_uuid	UUID, FK → Customer
total_amount	Total amount involved across all transactions in this case.	75000.0000	DECIMAL(20,4)
fraud_type	Category of fraud detected.	ACCOUNT_TAKEOVER	ENUM: ACCOUNT_TAKEOVER, CARD_FRAUD, UPI_FRAUD, IDENTITY_THEFT, PHISHING
investigation_notes	Analyst's investigation notes and findings.	Customer reported phone stolen...	String(4000), nullable
resolved_at	Timestamp of case resolution. Null while ongoing.	2026-05-15T16:00:00Z	Timestamp, nullable
outcome	Final outcome of the investigation.	FRAUD_CONFIRMED	ENUM: FRAUD_CONFIRMED, FALSE_POSITIVE, INSUFFICIENT_EVIDENCE

Entity: VelocityCounter

Redis-based sliding window counter for velocity rule evaluation. Tracks how many times a customer transacted in a recent time window in O(1).

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
key	Composite Redis key: customer_id:channel>window_seconds.	cust_uuid:UPI:600	String (Redis compound key)
count	Number of events in current window. Incremented atomically via Redis INCR.	4	Integer, Redis counter
amount_sum	Total rupee value transacted in window. Used for amount-based velocity rules.	150000.0000	DECIMAL, Redis field
window_start	Start of current sliding window.	2026-05-09T10:20:00Z	Timestamp, Redis field
window_end	End of current sliding window. After this, key expires.	2026-05-09T10:30:00Z	Timestamp, Redis TTL

Entity: BlockedEntity

Entities blocked from transacting: cards, accounts, devices, or IP addresses. Redis for fast per-request check; PostgreSQL for persistence and audit.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
entity_type	What type of entity is blocked.	CARD	ENUM: CARD, ACCOUNT, DEVICE, IP
entity_value	The identifier of the blocked entity (card_id, account_id, device_id, IP).	card_uuid	String, indexed
reason	Why this entity was blocked.	3 fraudulent transactions confirmed	String, not null

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
blocked_at	Timestamp of block.	2026-05-09T10:30:00Z	Timestamp, not null
blocked_by	Who blocked: SYSTEM or emp_id.	SYSTEM	String, not null
expires_at	Null = permanent. Non-null = auto-unblocked via TTL.	null	Timestamp, nullable

Entity: MIModelVersion

Tracks deployed ML model versions. Only one model version active at a time. Enables safe A/B testing and rollback if new model degrades performance.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
model_id	UUID PK.	model_uuid	UUID, PK
model_name	Name of the ML model.	FraudXGBoost	String, not null
version	Semantic version of this model.	2.3.1	String, not null
deployed_at	Timestamp when this version was deployed to production.	2026-05-01T02:00:00Z	Timestamp, not null
accuracy	Model accuracy on validation dataset.	0.9850	DECIMAL(5,4)
precision	Precision score — fraction of flagged transactions that were actual fraud.	0.9200	DECIMAL(5,4)
recall	Recall score — fraction of actual fraud correctly detected.	0.9450	DECIMAL(5,4)
active	Only one model version active at a time.	true	Boolean, unique constraint when true

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/fraud/evaluate/transaction	Synchronous risk evaluation (< 100ms SLA)
POST	/api/v1/fraud/evaluate/login	Evaluate login risk (called by Auth Service)
GET	/api/v1/fraud/alerts?status=	List fraud alerts for review (fraud team)
PUT	/api/v1/fraud/alerts/{alertId}/review	Review alert and mark as fraud/FP
POST	/api/v1/fraud/cases	Create fraud case from multiple alerts
GET	/api/v1/fraud/cases/{caseId}	Get fraud case details
GET	/api/v1/fraud/rules	List all active fraud rules
POST	/api/v1/fraud/rules	Create new fraud rule (DSL-based)
PUT	/api/v1/fraud/rules/{ruleId}	Update fraud rule threshold
DELETE	/api/v1/fraud/rules/{ruleId}	Deactivate a fraud rule
POST	/api/v1/fraud/blocklist	Block card/account/device/IP
DELETE	/api/v1/fraud/blocklist/{id}	Remove from blocklist
GET	/api/v1/fraud/customer/{customerId}/history	Fraud history for a customer
GET	/api/v1/fraud/metrics/dashboard	Real-time fraud metrics dashboard

Kafka Topics

Direction	Topic Name	Trigger / Consumer
PUBLISHES	fraud.transaction.blocked	Transaction + Payment Service — cancel the transaction
PUBLISHES	fraud.alert.created	Notification Service — alert customer + fraud team
SUBSCRIBES	transaction.initiated	Evaluate and block if fraudulent
SUBSCRIBES	payment.initiated	Pre-payment fraud scoring
SUBSCRIBES	auth.login.failed	Accumulate failed login signals
SUBSCRIBES	card.transaction.authorized	Card authorization scoring

Key Design Decisions

- Velocity counters: sliding window in Redis — e.g. '5 transactions in 10 minutes' evaluated in $O(1)$
- ML model served by separate FastAPI model server; called synchronously via Feign client with 80ms timeout
- Rule DSL: Drools rule engine embedded for complex CEP (Complex Event Processing) pattern matching
- Geolocation: MaxMind GeoIP2 database loaded in-memory — IP-to-location in <1ms with no external call
- False positive feedback loop: fraud analyst decisions → Kafka topic → model retraining pipeline (weekly)
- Device fingerprint mismatch: transaction from new device in new city for high amount → auto-OTP challenge

11 Reporting & Statement Service

Account statements · Tax certificates · Regulatory reports

Purpose & Responsibility

Generates all customer-facing and regulatory reports: Account statements (PDF/CSV, on-demand and scheduled), Interest Certificates (Section 80C TDS, Form 16A), Form 26AS reconciliation data, Fixed Deposit certificates, balance certificates for visa applications, Net Worth statements, CIBIL data correction reports, and regulatory submissions (RBI returns, SEBI reports, IT dept TDS returns). Implements CQRS read model with denormalised reporting tables fed by Kafka events.

Architectural Rationale

Reporting is read-heavy and computation-intensive — running complex SQL aggregations on transactional databases degrades OLTP performance. A separate service with a read-optimised PostgreSQL replica and pre-aggregated Cassandra tables ensures reports never impact live banking. S3 stores generated PDFs with pre-signed URLs for secure download.

Domain Entities

Entity: StatementRequest

Tracks an async account statement generation request. Customer requests a statement; system generates it asynchronously via Spring Batch and notifies when ready.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
request_id	UUID PK. Returned to client for polling.	req_uuid	UUID, PK
customer_id	FK to Customer who requested.	cust_uuid	UUID, FK → Customer
account_id	FK to Account for which statement is requested.	acc_uuid	UUID, FK → Account
from_date	Statement period start date.	2026-04-01	Date, not null
to_date	Statement period end date.	2026-04-30	Date, not null
format	Requested output format.	PDF	ENUM: PDF, CSV, EXCEL
status	Generation status.	COMPLETED	ENUM: QUEUED, PROCESSING, COMPLETED, FAILED
s3_path	S3 path to generated statement. Null until COMPLETED.	s3://nvsh-stmt/req_uuid.pdf	String, nullable
requested_at	Timestamp of request.	2026-05-01T10:00:00Z	Timestamp, not null
generated_at	Timestamp of completion. Null until COMPLETED.	2026-05-01T10:00:45Z	Timestamp, nullable

Entity: AccountSummaryView

Pre-aggregated monthly account summary. Materialised and refreshed every 15 minutes. Used for quick dashboard and reporting queries without hitting the OLTP database.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
account_id	FK to Account.	acc_uuid	UUID, FK → Account
customer_id	FK to Customer.	cust_uuid	UUID, FK → Customer
period	Year-month of this summary in YYYY-MM format.	2026-04	String(7), PK component with account_id
total_credits	Sum of all credits in this period.	50000.0000	DECIMAL(20,4)
total_debits	Sum of all debits in this period.	35000.0000	DECIMAL(20,4)
opening_balance	Balance at start of period.	80000.0000	DECIMAL(20,4)
closing_balance	Balance at end of period.	95000.0000	DECIMAL(20,4)
avg_balance	Average daily balance in this period. Used for minimum balance compliance.	87500.0000	DECIMAL(20,4)
updated_at	Timestamp of last materialisation.	2026-05-01T00:15:00Z	Timestamp, not null

Entity: TaxCertificate

Annual tax certificate for interest income (Form 16A for TDS deducted). Required by customers for income tax filing. Covers Indian financial year April–March.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
cert_id	UUID PK.	cert_uuid	UUID, PK
customer_id	FK to Customer.	cust_uuid	UUID, FK → Customer
account_id	FK to Account for which certificate is generated.	acc_uuid	UUID, FK → Account
financial_year	Indian financial year in YYYY-YY format.	2025-26	String(7), not null
interest_earned	Total interest credited during the financial year.	15000.0000	DECIMAL(20,4)
tds_deducted	Total TDS deducted on interest.	1500.0000	DECIMAL(20,4)
cert_type	Type of tax certificate.	FORM_16A	ENUM: FORM_16A, SECTION_80C, INTEREST_CERTIFICATE
s3_path	S3 path to generated PDF certificate.	s3://nvsh-taxcert/cert_uuid.pdf	String, not null
generated_at	Timestamp of certificate generation.	2026-04-01T06:00:00Z	Timestamp, not null

Entity: RegulatoryReport

Tracks submission of mandatory regulatory reports to RBI, SEBI, and Income Tax department. Submission status and acknowledgement references stored here.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
report_id	UUID PK.	report_uuid	UUID, PK
report_type	Type of regulatory report.	RBI_RETURN_BSR1	ENUM: RBI_RETURN_BSR1, CRILC, SEBI_QUARTERLY, TDS_26Q, etc.
reporting_period	Period covered in YYYY-MM format.	2026-04	String(7), not null
submitted_at	Timestamp of submission to regulator.	2026-05-10T10:00:00Z	Timestamp, nullable
status	Submission status.	SUBMITTED	ENUM: DRAFT, VALIDATED, SUBMITTED, ACKNOWLEDGED, REJECTED
file_path	S3 path to the report file.	s3://nvsh-regrep/report_uuid.xml	String, not null
submission_ref	Reference number issued by regulator upon acceptance.	RBI_ACK_2026050001	String, nullable

Entity: TxnAggregateDaily

Daily transaction aggregates per account stored in Cassandra. Fed by Kafka consumer processing transaction.completed events. Used for reporting without hitting OLTP.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
account_id	FK to Account. Cassandra partition key.	acc_uuid	UUID, Cassandra PK component
txn_date	Date of aggregation. Cassandra clustering key.	2026-05-09	Date, Cassandra clustering key
total_credits	Sum of credit transactions on this date.	25000.0000	DECIMAL(20,4)
total_debits	Sum of debit transactions on this date.	15000.0000	DECIMAL(20,4)
txn_count	Number of transactions on this date.	8	Integer
closing_balance	Account balance at end of this date.	95000.0000	DECIMAL(20,4)

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/reports/statement	Request account statement — async generation
GET	/api/v1/reports/statement/{requestId}	Poll statement generation status
GET	/api/v1/reports/statement/{requestId}/download	Signed S3 URL to download PDF/CSV
GET	/api/v1/reports/interest-certificate?fy=	Annual interest certificate (Section 80C)
GET	/api/v1/reports/tds-certificate?fy=	TDS certificate Form 16A
GET	/api/v1/reports/fd-certificate/{fdId}	FD receipt/certificate PDF
GET	/api/v1/reports/balance-certificate	Balance certificate for visa/loan applications

Method	Endpoint	Purpose
GET	/api/v1/reports/networth?customerId=	Customer net worth across all accounts
GET	/api/v1/admin/reports/rbi/returns?period=	RBI regulatory return data (compliance)
POST	/api/v1/admin/reports/rbi/submit	Submit RBI return report
GET	/api/v1/admin/reports/tds/annual?fy=	Annual TDS return data for IT dept filing

Kafka Topics

Direction	Topic Name	Trigger / Consumer
SUBSCRIBES	transaction.completed	Update TxnAggregateDaily for reporting
SUBSCRIBES	account.fd.matured	Generate FD maturity advice PDF
SUBSCRIBES	loan.disbursed	Update net worth report

Key Design Decisions

- Statement PDF generated using Apache PDFBox — letterhead, digital signature, QR code for verification
- Async generation: Spring Batch job spawned per request → PDF uploaded to S3 → customer notified
- Pre-signed S3 URL expires in 2 hours — customer must re-request after expiry for security
- Materialised views refreshed every 15 minutes via Spring Scheduler — reporting never hits OLTP DB
- Tax year = April to March (Indian FY); all certificates support multi-year data

12 Audit & Compliance Service

Immutable audit trail · RBI compliance · AML · FATCA · PEP/Sanctions

Purpose & Responsibility

Maintains an immutable, tamper-evident audit trail of every system action: who did what, when, from where, and what changed. Implements AML (Anti-Money Laundering) transaction monitoring per RBI/PMLA guidelines, FATCA/CRS compliance for NRI accounts, PEP (Politically Exposed Person) screening, sanctions screening (OFAC/UN/RBI lists), suspicious transaction reporting (STR) to FIU-IND, and regulatory examination support (read-only access for RBI auditors).

Architectural Rationale

Audit is a regulatory mandate — RBI requires 10-year audit trail retention. A separate service ensures: audit data is append-only (no UPDATE/DELETE), audit records are cryptographically signed (each event hashes previous event — blockchain-inspired), Elasticsearch enables full-text search across billions of audit events, and AML rules run independently without impacting transaction speed.

Domain Entities

Entity: AuditEvent

Immutable record of every system action. Append-only — DB user has no DELETE/UPDATE privilege. SHA-256 hash chain links events: tamper of any event breaks the chain.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
event_id	UUID PK. Append-only — never updated or deleted.	event_uuid	UUID, PK, append-only
service_name	Which microservice generated this event.	transaction-service	String, not null
action	What action was performed.	TRANSACTION_POSTED	String, not null
entity_type	Type of entity the action was performed on.	TRANSACTION	String e.g. TRANSACTION, ACCOUNT, CUSTOMER, CARD
entity_id	ID of the specific entity.	txn_uuid	UUID, not null, indexed
customer_id	Which customer's data was affected. Used for customer-specific audit queries.	cust_uuid	UUID, nullable, indexed
user_id	Who performed the action. Service-generated actions have a service account user_id.	user_uuid	UUID, not null
old_value_json	Previous state of the entity before change. Null for create operations.	{"status":"PENDING"}	JSON, nullable
new_value_json	New state of the entity after change.	{"status":"POSTED"}	JSON, nullable
ip_address	IP address from which the action originated.	203.0.113.42	String(45)
timestamp	Exact timestamp with microsecond precision.	2026-05-09T10:30:00.123456Z	Timestamp(6), not null, indexed
hash	SHA-256 hash: SHA256(event_data + previous_event_hash). Tamper-evident chain.	sha256_hash_64_chars	String(64), not null

Entity: AmlAlert

Created when an AML rule fires during transaction monitoring. Examples: structuring (multiple transactions just below INR 10L reporting threshold), round-trip transactions.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
alert_id	UUID PK.	aml_alert_uuid	UUID, PK
customer_id	FK to Customer whose transactions triggered the alert.	cust_uuid	UUID, FK → Customer
rule_triggered	Which AML rule triggered this alert.	STRUCTURING_DETECTION	String, FK → ComplianceRule.rule_id
transaction_ids	Array of transaction IDs forming the suspicious pattern.	["txn1", "txn2", "txn3"]	UUID[], not null
amount	Total amount involved in the suspicious pattern.	950000.0000	DECIMAL(20,4)
pattern	Description of suspicious pattern detected.	9 transactions of INR 99,000 each in 24 hours	String, not null

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
status	Review status.	OPEN	ENUM: OPEN, UNDER_REVIEW, STR_FILED, CLOSED_NO_ACTION
assigned_analyst	Compliance analyst employee ID assigned for review.	emp_00890	String, nullable
created_at	Timestamp of alert creation.	2026-05-09T10:30:00Z	Timestamp, not null

Entity: SuspiciousTransactionReport

Mandatory report filed with FIU-IND (Financial Intelligence Unit — India) when a transaction is suspected to involve money laundering or terrorist financing.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
str_id	UUID PK.	str_uuid	UUID, PK
customer_id	FK to Customer being reported.	cust_uuid	UUID, FK → Customer
alert_id	FK to AmlAlert that triggered this STR.	aml_alert_uuid	UUID, FK → AmlAlert
amount	Total amount involved in suspicious activity.	950000.0000	DECIMAL(20,4)
nature_of_transaction	Description of suspicious activity per FIU-IND format requirements.	Multiple structured cash deposits below reporting threshold	String(2000), not null
filed_at	Timestamp when STR was submitted to FIU-IND via SFTP.	2026-05-10T10:00:00Z	Timestamp, not null
fiu_ack_number	FIU-IND acknowledgement reference. Null until FIU confirms receipt.	FIU2026051000001	String, nullable
status	STR filing status.	FILED	ENUM: DRAFT, FILED, ACKNOWLEDGED, SUPPLEMENTARY_FILED

Entity: PepRecord

Database of Politically Exposed Persons. Customers matching PEP records are subject to Enhanced Due Diligence (EDD) per RBI's KYC Master Direction.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
pep_id	UUID PK.	pep_uuid	UUID, PK
name	Full name of the PEP. Fuzzy matching used during screening.	Example PEP Name	String, not null
dob	Date of birth for disambiguation of common names.	1950-09-17	Date, nullable
nationality	Nationality of the PEP.	IND	String(3), ISO 3166-1 alpha-3
position	Current or former political/government position.	Former Chief Minister	String, not null
risk_level	Risk classification of this PEP.	HIGH	ENUM: MEDIUM, HIGH, VERY_HIGH
source	Data source for this PEP record.	WORLD_CHECK	ENUM: WORLD_CHECK, REFINITIV, RBI_LIST, INTERNAL
added_at	Timestamp when this record was added to the database.	2026-01-01T00:00:00Z	Timestamp, not null

Entity: SanctionsList

Database of sanctioned entities from OFAC, UN, and RBI lists. Bloom filter in Redis enables $O(1)$ negative check before expensive DB lookup.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
sanction_id	UUID PK.	sanction_uuid	UUID, PK
entity_name	Primary name of the sanctioned entity.	Sanctioned Entity Name	String, not null
aliases	List of known aliases/alternate names. Used in fuzzy matching.	["Alias1", "Alias2"]	String[], not null
country	Country of the sanctioned entity.	SY	String(2), ISO 3166-1 alpha-2
sanction_type	Category of sanction.	SDN	ENUM: SDN, SECTORAL, CONSOLIDATED, RBI_CAUTION
list_source	Which sanctions list this record comes from.	OFAC	ENUM: OFAC, UN_CONSOLIDATED, RBI, EU

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
added_at	Timestamp when record was added. Lists refreshed daily.	2026-01-15T06:00:00Z	Timestamp, not null

Entity: CustomerScreening

Result of PEP and sanctions screening for a customer. Performed at KYC time and periodically thereafter. cleared_by required for any match before account activation.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
screening_id	UUID PK.	screening_uuid	UUID, PK
customer_id	FK to Customer being screened.	cust_uuid	UUID, FK → Customer, indexed
pep_match	True if customer's name/DOB matched a PEP record.	false	Boolean, not null
sanctions_match	True if customer matched a sanctioned entity.	false	Boolean, not null
match_score	Fuzzy match confidence score 0-100. Scores above 80 require manual review.	0	Integer(0-100)
screened_at	Timestamp of screening.	2026-01-20T10:00:00Z	Timestamp, not null
cleared_by	Employee ID who cleared a false-positive match. Null if no match or pending.	null	String, nullable

Entity: ComplianceRule

Configurable AML/KYC/FATCA rule. Defines thresholds and actions for automated compliance monitoring.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
rule_id	UUID PK.	crule_uuid	UUID, PK
category	Compliance domain this rule belongs to.	AML	ENUM: AML, KYC, FATCA, CRS, PEP_SCREENING
description	Human-readable description of what this rule detects.	Cash transactions above INR 10L threshold	String, not null
threshold	Numeric threshold that triggers the rule.	1000000.00	DECIMAL(20,4), nullable
action	What happens when rule triggers.	CREATE_AML_ALERT	ENUM: CREATE_AML_ALERT, FILE_CTR, REQUIRE_EDD, BLOCK_TRANSACTION
active	Rules can be deactivated without deletion.	true	Boolean, default: true

REST API Endpoints

Method	Endpoint	Purpose
GET	/api/v1/audit/events?entityId=&from;=&to;=	Search audit events by entity and date range
GET	/api/v1/audit/events/{eventId}	Get specific audit event with hash proof
GET	/api/v1/audit/customer/{customerId}	All audit events for a customer (RBI examiner)
GET	/api/v1/audit/verify/{eventId}	Verify audit event hash chain integrity
GET	/api/v1/compliance/aml/alerts?status=	List AML alerts for analyst review
PUT	/api/v1/compliance/aml/alerts/{id}/review	Review AML alert decision
POST	/api/v1/compliance/str/file	File STR with FIU-IND (auto-generates XML)
GET	/api/v1/compliance/str/{strId}	Get STR status and FIU acknowledgement
POST	/api/v1/compliance/screening/customer	Screen customer against PEP/sanctions lists
GET	/api/v1/compliance/sanctions/check?name=	Ad-hoc sanctions check for a name
GET	/api/v1/compliance/fatca/customers	FATCA-reportable customer list (US persons)
GET	/api/v1/compliance/rules	List active compliance rules
POST	/api/v1/compliance/rules	Create compliance rule

Kafka Topics

Direction	Topic Name	Trigger / Consumer
SUBSCRIBES	gateway.request.logged	Archive all API requests as audit events
SUBSCRIBES	transaction.completed	AML screening of every transaction
SUBSCRIBES	customer.profile.updated	Audit customer data changes
SUBSCRIBES	auth.password.changed	Security audit event
SUBSCRIBES	card.blocked	Security event audit

Key Design Decisions

- Audit events are append-only — DB user has no DELETE/UPDATE privilege on audit tables
- SHA-256 hash chain: `event_hash = SHA256(event_data + previous_event_hash)` — tamper evident
- Elasticsearch index per year with 10-year retention policy — 100% searchable audit archive
- AML rules: structuring detection (multiple txns just below INR 10L threshold), round-trip detection
- STR XML auto-generated in FIU-IND prescribed format (FMR-1, FMR-2) and submitted via SFTP
- PEP/Sanctions lists refreshed daily from OFAC, UN Consolidated List, RBI notification via scheduled batch

13 Investment & Wealth Service

Mutual Funds · Demat · Insurance · Sovereign Gold Bonds · NPS

Purpose & Responsibility

Third-party investment product distribution: Mutual Fund investments (via BSE StAR MF / MFU platform), Demat account opening and equity trading referral (linked to partner broker), Sovereign Gold Bonds (SGB) subscription during RBI issuance windows, NPS (National Pension System) account opening and contribution, Life and General Insurance premium collection (as Corporate Agent), and Portfolio tracking with returns calculation. This service is a distributor, not a product manufacturer.

Architectural Rationale

Wealth management is a growing revenue stream for banks (commission-based). A separate service allows: SEBI compliance maintained independently (distributor regulations), integration with multiple AMC APIs without touching core banking, and the service can be completely disabled if regulatory environment changes without affecting savings/loans/payments.

Domain Entities

Entity: InvestmentAccount

Master investment account for a customer. A customer can have separate accounts for MF, Demat, and NPS. Status controls whether transactions are allowed.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
inv_account_id	UUID PK.	inv_uuid	UUID, PK
customer_id	FK to Customer who owns this investment account.	cust_uuid	UUID, FK → Customer
type	Type of investment account.	MF	ENUM: MF, DEMAT, NPS
status	Account status.	ACTIVE	ENUM: ACTIVE, SUSPENDED, CLOSED
created_at	Timestamp of account opening.	2026-02-01T10:00:00Z	Timestamp, not null
kyc_ref	Reference to the KYC record used. SEBI requires separate KYC for investments.	kyc_uuid	UUID, FK → KycDocument

Entity: MutualFundHolding

Current mutual fund holding for a customer. Units, NAV, and current value updated daily when AMFI publishes NAVs at 10 PM.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
holding_id	UUID PK.	holding_uuid	UUID, PK
inv_account_id	FK to InvestmentAccount.	inv_uuid	UUID, FK → InvestmentAccount
scheme_code	AMFI scheme code. Unique identifier for the MF scheme.	120503	String, not null, indexed
scheme_name	Full scheme name.	Axis Bluechip Fund — Direct Growth	String, not null
units	Number of units held. Updated after each purchase and redemption.	1250.500	DECIMAL(15,3)
purchase_nav	Average NAV at which units were purchased (Average Cost basis).	45.2340	DECIMAL(10,4)
current_nav	Latest NAV from AMFI. Updated daily at 10 PM.	52.1560	DECIMAL(10,4)
invested_amount	Total amount invested in this scheme.	56542.5000	DECIMAL(20,4)
current_value	Current market value = units * current_nav.	65215.7000	DECIMAL(20,4), derived
last_updated	Timestamp of last NAV update.	2026-05-09T22:00:00Z	Timestamp, not null

Entity: SipInstruction

Systematic Investment Plan — automatic monthly MF purchase of a fixed amount on a set date. Executed by the Payment Service via NACH mandate or direct debit.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
sip_id	UUID PK.	sip_uuid	UUID, PK

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
inv_account_id	FK to InvestmentAccount.	inv_uuid	UUID, FK → InvestmentAccount
scheme_code	AMFI scheme code of the target scheme.	120503	String, not null
amount	Fixed monthly SIP amount.	5000.0000	DECIMAL(20,4), check >= 500 (SEBI minimum)
frequency	SIP frequency.	MONTHLY	ENUM: WEEKLY, MONTHLY, QUARTERLY
start_date	Date of first SIP installment.	2026-02-01	Date, not null
end_date	Date of last installment. Null = perpetual SIP.	2031-02-01	Date, nullable
next_debit_date	Date of next installment. Updated after each execution.	2026-06-01	Date, not null
status	SIP lifecycle state.	ACTIVE	ENUM: ACTIVE, PAUSED, CANCELLED, COMPLETED

Entity: GoldBondHolding

Sovereign Gold Bonds (SGB) issued by RBI on behalf of Government of India. Backed by physical gold. Earns 2.5% annual interest plus capital appreciation.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
bond_id	UUID PK.	bond_uuid	UUID, PK
inv_account_id	FK to InvestmentAccount.	inv_uuid	UUID, FK → InvestmentAccount
series	RBI SGB series identifier.	SGB 2025-26 Series I	String, not null
quantity_grams	Quantity of gold in grams this bond represents. Minimum 1 gram.	10.000	DECIMAL(10,3), check >= 1
issue_price	Price per gram paid at time of subscription.	6500.0000	DECIMAL(10,4), per gram, INR
current_price	Current gold price per gram from RBI. Updated daily.	7200.0000	DECIMAL(10,4), per gram, INR
maturity_date	SGB matures after 8 years. Tax-free redemption at maturity.	2033-10-30	Date, not null

Entity: InsurancePolicy

Insurance policy distributed by Nivesh Bank as a Corporate Agent. Bank collects premiums and earns commission. Not the insurer — just the distributor.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
policy_id	UUID PK.	policy_uuid	UUID, PK
customer_id	FK to Customer who holds the policy.	cust_uuid	UUID, FK → Customer
insurer_name	Name of the insurance company.	LIC of India	String, not null
policy_number	Policy number issued by the insurer.	LIC-2026-001234567	String, unique, not null
product_type	Type of insurance product.	TERM_LIFE	ENUM: TERM_LIFE, ENDOWMENT, ULIP, HEALTH, MOTOR, TRAVEL
premium_amount	Premium amount per payment frequency.	12000.0000	DECIMAL(20,4)
premium_frequency	How often premiums are paid.	ANNUAL	ENUM: MONTHLY, QUARTERLY, HALF_YEARLY, ANNUAL
next_premium_date	Date of next premium due.	2027-01-15	Date, not null
sum_assured	Insurance coverage amount.	10000000.0000	DECIMAL(20,4)
status	Policy status.	ACTIVE	ENUM: ACTIVE, LAPSED, SURRENDERED, MATURED, CLAIMED

Entity: NpsAccount

National Pension System account. PRAN (Permanent Retirement Account Number) is the unique identifier issued by NSDL CRA. Tier I is mandatory pension; Tier II is voluntary.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
nps_id	UUID PK.	nps_uuid	UUID, PK

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
customer_id	FK to Customer.	cust_uuid	UUID, FK → Customer
pran_number	Permanent Retirement Account Number issued by NSDL CRA. Unique across India.	1101234567890	String(13), unique, not null
tier	NPS tier type. Tier I: mandatory with tax benefits. Tier II: voluntary withdrawal anytime.	I	ENUM: I, II
equity_allocation	Percentage allocated to equity (E) funds. Max 75% until age 50.	75.00	DECIMAL(5,2), percent
debt_allocation	Percentage allocated to government bonds (G) funds.	25.00	DECIMAL(5,2), percent
current_value	Current market value of NPS corpus based on latest NAV.	350000.0000	DECIMAL(20,4)

Entity: MarketDataCache

Redis cache for mutual fund NAV data fetched from AMFI API daily at 10 PM. 24-hour TTL ensures stale data is never served.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
scheme_code	AMFI scheme code. Redis key.	120503	String, Redis key
nav	Latest NAV of the scheme.	52.1560	DECIMAL(10,4), Redis field
nav_date	Date for which this NAV was published.	2026-05-09	Date, Redis field
fund_house	Name of the Asset Management Company.	Axis Mutual Fund	String, Redis field
category	SEBI-defined category of the scheme.	Large Cap	String, Redis field
1y_return	1-year annualised return percentage.	18.50	DECIMAL(6,2), percent
3y_return	3-year annualised return percentage.	22.30	DECIMAL(6,2), percent
5y_return	5-year annualised return percentage.	19.80	DECIMAL(6,2), percent

REST API Endpoints

Method	Endpoint	Purpose
POST	/api/v1/investments/mf/account	Open mutual fund investment account
GET	/api/v1/investments/mf/schemes	Browse mutual fund schemes with filters
GET	/api/v1/investments/mf/schemes/{code}	Scheme details with NAV and returns
POST	/api/v1/investments/mf/purchase	Lump-sum MF purchase (debit savings account)
POST	/api/v1/investments/mf/sip	Start a SIP instruction
PUT	/api/v1/investments/mf/sip/{sipId}	Pause/modify/cancel SIP
POST	/api/v1/investments/mf/redeem	Redeem MF units (credit to savings account)
GET	/api/v1/investments/portfolio?customerId=	Full investment portfolio with current values
GET	/api/v1/investments/portfolio/returns	XIRR returns across all holdings
POST	/api/v1/investments/sgb/subscribe	Subscribe to Sovereign Gold Bond (during window)
GET	/api/v1/investments/sgb/holdings	SGB holding details
POST	/api/v1/investments/nps/account	Open NPS account and generate PRAN
POST	/api/v1/investments/nps/contribute	NPS contribution via savings account debit
GET	/api/v1/investments/nps/{npsId}	NPS account statement and balance
POST	/api/v1/investments/insurance/premium/pay	Pay insurance premium
GET	/api/v1/investments/insurance?customerId=	List insurance policies

Kafka Topics

Direction	Topic Name	Trigger / Consumer
PUBLISHES	investment.mf.purchased	Transaction Service — debit savings account
PUBLISHES	investment.sip.due	Payment Service — trigger SIP debit

Direction	Topic Name	Trigger / Consumer
PUBLISHES	investment.insurance.premium.due	Notification Service — premium reminder

Key Design Decisions

- MF transactions go via BSE StAR MF API — order placed, confirmation received async via webhook
- NAV data fetched from AMFI API daily at 10 PM; cached in Redis for real-time portfolio valuation
- XIRR calculation implemented using Newton-Raphson method in Java — exact financial math
- SGB subscription only active during RBI issuance windows; service checks window via RBI API
- AMFI compliance: distributor code (ARN) attached to every MF transaction — regulatory requirement

14 Branch & ATM Management Service

Branch network · ATM fleet · Appointment booking · Geo-search

Purpose & Responsibility

Manages the physical network: 22,000+ branches and 65,000+ ATMs (SBI-scale). Branch management (working hours, services offered, staff, address, geolocation), ATM fleet management (cash level monitoring, status, maintenance scheduling), customer appointment booking for branches, ATM cash replenishment scheduling, ATM health monitoring and incident management, ATM software/hardware inventory, and locator APIs for the mobile app (nearest branch/ATM with services filter).

Architectural Rationale

Branch and ATM operations are operational systems distinct from financial data. A dedicated service allows field teams to manage branch/ATM data without banking system access. MongoDB's geospatial indexing (2dsphere) enables sub-10ms nearest-branch queries across 22,000+ locations. ATM status feeds into the Fraud Detection service to flag transactions from out-of-service ATMs.

Domain Entities

Entity: Branch

A Nivesh Bank branch. Includes geolocation for proximity searches. IFSC code uniquely identifies the branch for NEFT/RTGS routing. Only 1 branch per 5km radius enforced at onboarding.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
branch_id	UUID PK.	branch_uuid	UUID, PK
ifsc_code	11-char IFSC: NVSH0 + PIN_ZONE(3) + SEQ(3). Unique. Used as routing code in all fund transfers.	NVSH0400001	String(11), unique, not null
name	Branch name for display in mobile app and statements.	Nivesh Bank — Bandra West Branch	String, not null
address	Full branch address.	101 Hill Road, Bandra West, Mumbai	String, not null
geolocation	Latitude/longitude for proximity searches. 2dsphere index in MongoDB.	{"lat":19.0596,"lng":72.8295}	GeoJSON Point, 2dsphere indexed in MongoDB
working_hours_json	JSON defining working hours per day of week.	{"MON-FRI":"09:00-17:00","SAT":"09:00-13:00"}	JSON, not null
services_json	Array of services available at this branch.	["ACCOUNT_OPENING","LOANS","FOREX","SAFE_DEPOSIT"]	JSON array
manager_name	Name of the branch manager.	Anita Desai	String, not null
phone	Branch phone number for customer calls.	+91-22-26401234	String, not null
email	Branch email address.	bandra.west@nivesh.bank	String, not null
status	Whether branch is operational.	OPERATIONAL	ENUM: OPERATIONAL, TEMPORARILY_CLOSED, PERMANENTLY_CLOSED, UNDER_RENOVATION
zone	Geographic zone for regional reporting.	WEST	ENUM: NORTH, SOUTH, EAST, WEST, CENTRAL, NORTHEAST
state	Indian state where branch is located.	Maharashtra	String, not null

Entity: AtmTerminal

An ATM terminal operated by Nivesh Bank. Cash level monitored every 15 minutes via controller webhook. Status published to Kafka for fraud detection and customer app.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
atm_id	UUID PK.	atm_uuid	UUID, PK
terminal_id	Physical terminal ID printed on the ATM. Used by field engineers.	ATM-MUM-001234	String, unique, not null
branch_id	FK to Branch if ATM is on branch premises. Null for off-site ATMs.	branch_uuid	UUID, nullable FK → Branch

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
location_address	Full address of ATM location for customer-facing locator.	Juhu Beach Promenade, Mumbai	String, not null
geolocation	Latitude/longitude for proximity search. 2dsphere indexed in MongoDB.	{"lat":19.0944,"lng":72.8264}	GeoJSON Point, 2dsphere indexed
cash_level_percent	Current cash fill level as percentage of ATM's cassette capacity.	65	Integer(0-100), updated every 15 min
status	Current operational status.	ONLINE	ENUM: ONLINE, OFFLINE, MAINTENANCE, CASH_DEPLETED, OUT_OF_ORDER
vendor	ATM hardware vendor.	NCR	ENUM: NCR, DIEBOLD_NIXDORF, WINCOR, GRG
model	ATM hardware model.	NCR SelfServ 87	String, not null
last_cash_refill	Timestamp of last cash replenishment.	2026-05-08T09:00:00Z	Timestamp, not null
software_version	ATM software version. Used for patch management.	v4.2.1	String, not null

Entity: AtmCashLog

Log of every cash refill event. Used for cash management analytics and forecasting. Helps determine optimal cash replenishment schedules.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
log_id	UUID PK.	atm_log_uuid	UUID, PK
atm_id	FK to AtmTerminal.	atm_uuid	UUID, FK → AtmTerminal
cash_level_before	Cash level percentage before refill.	10	Integer(0-100)
cash_level_after	Cash level percentage after refill.	100	Integer(0-100)
refill_amount	Amount of cash loaded in INR.	2000000.0000	DECIMAL(20,4)
refill_by	Name of the cash van agency that performed the refill.	Brinks India	String, not null
logged_at	Timestamp of refill.	2026-05-08T09:00:00Z	Timestamp, not null

Entity: AtmIncident

Tracks ATM faults and incidents reported by customers or detected by monitoring. Assigned to vendor for resolution. Used for ATM SLA management.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
incident_id	UUID PK.	incident_uuid	UUID, PK
atm_id	FK to AtmTerminal.	atm_uuid	UUID, FK → AtmTerminal
incident_type	Type of fault.	CARD_READER_FAILURE	ENUM: CARD_READER_FAILURE, CASH_JAM, NETWORK_FAILURE, DISPLAY_FAILURE, VANDALISM, POWER_FAILURE
reported_at	Timestamp when incident was reported or detected.	2026-05-09T10:00:00Z	Timestamp, not null
severity	Severity of the incident. CRITICAL = ATM completely non-functional.	HIGH	ENUM: LOW, MEDIUM, HIGH, CRITICAL
assigned_vendor	Vendor assigned for repair.	NCR Services	String, nullable
resolved_at	Timestamp of resolution. Null while open.	2026-05-09T14:00:00Z	Timestamp, nullable
root_cause	Root cause analysis after resolution.	Card reader magnetic head worn out	String, nullable

Entity: BranchAppointment

Customer appointment booking for specific services at a branch. Reduces wait times. Redis slot locking prevents double-booking. Token number for queue management.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
appt_id	UUID PK.	appt_uuid	UUID, PK
customer_id	FK to Customer who booked.	cust_uuid	UUID, FK → Customer

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
branch_id	FK to Branch where appointment is booked.	branch_uuid	UUID, FK → Branch
service_type	Which service the customer is coming for.	HOME_LOAN_CONSULTATION	String, FK → ServiceDefinition
slot_date	Date of appointment.	2026-05-15	Date, not null
slot_time	Time slot of appointment.	10:30	Time, not null
status	Appointment status.	CONFIRMED	ENUM: CONFIRMED, CANCELLED, COMPLETED, NO_SHOW
token_number	Queue token number. Displayed on screen at branch.	A-042	String, nullable — assigned on day of visit
notes	Customer's notes about what they need.	Need to discuss home loan pre-approval	String, nullable

Entity: ServiceDefinition

Defines the services offered at branches. Controls whether appointment is mandatory and estimated service duration for slot calculation.

Field Name	Purpose & Business Meaning	Example Value	Type / Enum Values
service_id	UUID PK.	service_uuid	UUID, PK
name	Service name.	Home Loan Consultation	String, unique, not null
available_at_branch_types	List of branch types where this service is offered. Not all services at all branches.	["FULL_SERVICE", "PREMIUM", "FLAGSHIP"]	String[], not null
requires_appointment	True if walk-in not allowed — appointment mandatory.	true	Boolean, default: false
estimated_duration_min	Estimated duration in minutes. Used for slot capacity planning.	45	Integer, not null

REST API Endpoints

Method	Endpoint	Purpose
GET	/api/v1/branches/search?lat=&lng;=&radius;=&service;=	Find nearest branches with service filter
GET	/api/v1/branches/{branchId}	Branch details including working hours
GET	/api/v1/branches/{branchId}/services	Services available at this branch
POST	/api/v1/branches/{branchId}/appointments	Book appointment at a branch
GET	/api/v1/branches/{branchId}/appointments/slots?date=	Available appointment slots
GET	/api/v1/appointments/{apptId}	Get appointment details and token
PUT	/api/v1/appointments/{apptId}/cancel	Cancel an appointment
GET	/api/v1/atms/search?lat=&lng;=&radius;=	Find nearest ATMs with cash availability
GET	/api/v1/atms/{atmId}	ATM details and current status
GET	/api/v1/atms/{atmId}/cash-level	Current cash level percentage (ATM ops)
PUT	/api/v1/atms/{atmId}/status	Update ATM status (ops team)
POST	/api/v1/atms/{atmId}/incidents	Report ATM incident
GET	/api/v1/admin/atms/cash-alerts	ATMs below cash threshold (ops dashboard)
GET	/api/v1/admin/branches	All branches list with status (admin)
PUT	/api/v1/admin/branches/{branchId}	Update branch information

Kafka Topics

Direction	Topic Name	Trigger / Consumer
PUBLISHES	atm.status.changed	Fraud Detection — flag txns from offline ATMs
PUBLISHES	atm.cash.low	Operations team alert for cash replenishment
PUBLISHES	branch.appointment.confirmed	Notification Service — appointment confirmation SMS

Key Design Decisions

- MongoDB 2dsphere index on geolocation field — \$nearSphere query returns nearest in <5ms for 65,000 ATMs
- ATM cash level reported by ATM controller software via REST webhook every 15 minutes
- Appointment booking uses Redis-based slot locking — prevents double booking within 5-second window
- ATM offline status published to Kafka → Fraud Detection ignores transactions from offline ATMs
- Holiday calendar integrated — branch working hours override for national/state holidays
- 5km exclusivity rule: PostGIS ST_DWithin check prevents branch registration if existing branch nearby

21 Inter-Service Communication & Event Catalog

All asynchronous events flow through Apache Kafka (3 brokers, RF=3, min.insync.replicas=2). Topics partitioned by customer_id for per-customer ordering guarantees. Confluent Schema Registry enforces Avro schemas — no schema drift possible.

Complete Kafka Topic Registry

Topic Name	Producer	Consumers	Parts	Retention
auth.user.registered	Auth Service	Customer Profile	12	7 days
auth.login.failed	Auth Service	Fraud Detection	12	7 days
auth.account.locked	Auth Service	Notification, Audit	12	30 days
customer.kyc.completed	Customer Profile	Account Mgmt	12	30 days
customer.profile.updated	Customer Profile	Notification, Audit	12	30 days
account.opened	Account Mgmt	Notification, Audit	24	30 days
account.dormant	Account Mgmt	Notification, Compliance	12	90 days
account.fd.matured	Account Mgmt	Transaction, Notification	12	30 days
transaction.initiated	Transaction Svc	Fraud Detection	48	3 days
transaction.completed	Transaction Svc	Account, Notification, Audit, Reporting	48	30 days
transaction.failed	Transaction Svc	Notification, Audit	24	30 days
transaction.reversed	Transaction Svc	Account, Notification, Audit	24	30 days
payment.initiated	Payments Svc	Fraud Detection	24	3 days
payment.completed	Payments Svc	Transaction, Notification, Audit	24	30 days
payment.failed	Payments Svc	Notification, Audit	24	30 days
fraud.transaction.blocked	Fraud Detection	Transaction, Payment, Notification	24	90 days
fraud.alert.created	Fraud Detection	Notification, Audit	24	90 days
card.transaction.authorized	Card Mgmt	Fraud Detection	24	3 days
card.blocked	Card Mgmt	Notification, Audit	12	90 days
loan.disbursed	Loan Service	Transaction, Notification, Audit	12	365 days
loan.emi.overdue	Loan Service	Notification, Fraud, Compliance	12	365 days

Synchronous Inter-Service REST Calls (Spring OpenFeign)

Caller	Called Service	Endpoint	When
Transaction Svc	Account Svc	POST /internal/accounts/{id}/debit	Before posting debit
Transaction Svc	Account Svc	POST /internal/accounts/{id}/credit	After confirming debit
Transaction Svc	Fraud Detection	POST /fraud/evaluate/transaction	Pre-authorisation check
Payment Svc	Account Svc	GET /accounts/{id}/balance	Pre-payment balance check
Payment Svc	Customer Profile	GET /customers/{id}	Beneficiary name verification
Auth Service	Fraud Detection	POST /fraud/evaluate/login	Login risk scoring
Notification Svc	Customer Profile	GET /customers/{id}/preferences	Channel preference lookup
Reporting Svc	Account Svc	GET /accounts/{id}/mini-statement	Mini-statement data fetch
Card Svc	Account Svc	GET /accounts/{id}/balance	Credit limit check
Loan Svc	Customer Profile	GET /customers/{id}/kyc/status	Eligibility check

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Security Architecture

Layer	Control	Implementation
Network	VPC, Security Groups, WAF	AWS WAF blocks OWASP Top 10; SGs whitelist only API GW traffic
Transport	TLS 1.3 everywhere	No TLS 1.2; certificate pinning in mobile app; mTLS between services
Authentication	OAuth2 + JWT + MFA	Spring Authorization Server; 15-min token expiry; MFA enforced for transfers
Authorization	RBAC + ABAC	Spring Security @PreAuthorize; attribute-based for amount/account ownership
Data at Rest	AES-256-GCM	AWS KMS-managed keys; Vault for field-level encryption (PAN, Aadhaar)
Data in Transit	TLS 1.3	All S3, RDS, ElastiCache connections encrypted in transit
Secrets	HashiCorp Vault	No .env files; dynamic DB credentials rotated every 24 hours
API Security	Rate limiting, CORS, CSRF	Spring Cloud Gateway enforces; CSRF tokens for web sessions
Intrusion Detection	AWS GuardDuty + CloudTrail	Real-time threat detection; all API calls logged to CloudTrail
Pen Testing	Quarterly VAPT	CERT-In empanelled agency per RBI IT Framework
PCI-DSS	Level 1 Compliance	Annual QSA audit; cardholder data scoped to Card Service only

23 Data Architecture & Database Strategy

Service	Primary DB	Rationale	Replication	Backup
API Gateway	Redis Cluster	Rate limit counters, route cache — pure in-memory speed	3 nodes	RDB snapshot hourly
Auth Service	PostgreSQL 16	ACID for user/token tables; concurrent login safety	1 primary + 2 replicas	Continuous WAL archiving
Customer Profile	PostgreSQL + MongoDB	Structured KYC in PG; unstructured video KYC in Mongo	2 replicas each	Point-in-time recovery
Account Mgmt	PostgreSQL 16	DECIMAL precision for balances; ACID transactions	1 primary + 3 replicas (multi-AZ)	RPO < 5 sec
Transaction Proc	PostgreSQL (hot) + Cassandra (cold)	Hot: current month OLTP; Cold: 5yr history time-series	Sync replication	Cassandra RF=3
Payments	PostgreSQL 16	Saga state, payment orders, mandates — ACID required	2 replicas	PITR 30 days
Loan Service	PostgreSQL 16	Complex FKs (EMI→Loan→Application), financial calcs	2 replicas	PITR 30 days
Card Service	PostgreSQL 16	PCI-DSS: isolated DB, encrypted tablespace	2 replicas in PCI zone	Encrypted backup
Notification	MongoDB	Flexible schema for varied notification payloads	Replica set 3 nodes	Daily snapshot
Fraud Detection	Redis + Cassandra	Redis: velocity counters (ms latency); Cassandra: risk history	Redis Cluster; Cassandra RF=3	Cassandra snapshots
Reporting	PostgreSQL read replica + S3	Materialised views on read replica; PDFs on S3	Read replica from Txn DB	S3 versioning 10yr
Audit	PostgreSQL + Elasticsearch	PG: structured audit; ES: full-text search across events	ES 5-node cluster	10-year retention
Investment	PostgreSQL	MF/SGB/NPS accounts — relational products	2 replicas	PITR 30 days
Branch & ATM	PostgreSQL + MongoDB	PG: structured; MongoDB: geospatial 2dsphere indexes	1 replica each	Daily backup

24 Deployment & DevOps Strategy

Concern	Tool/Technology	Details
Container Runtime	Docker + Distroless images	Minimal attack surface; non-root user; no shell in prod image
Orchestration	Kubernetes AWS EKS 1.29	Managed K8s; Karpenter node autoscaling; pod HPA on CPU + custom metrics
Service Mesh	Istio	mTLS between pods; traffic shifting for canary; circuit breaking; distributed tracing
CI/CD Pipeline	Jenkins + ArgoCD	Jenkins: build, test, scan, push image; ArgoCD: GitOps deployment to K8s
Code Quality	SonarQube + Checkstyle	Quality gate: 0 critical bugs, 80% coverage, no code smells blocking merge
Security Scanning	Snyk + Trivy	Snyk: dependency vulnerabilities; Trivy: container image; fail pipeline on HIGH
IaC	Terraform + Helm	All AWS resources in Terraform; K8s manifests in Helm charts
Config Mgmt	Spring Cloud Config + Vault	Git-backed configs; Vault for secrets injection at pod startup
Metrics	Prometheus + Grafana	Micrometer exposes metrics; RED dashboards (Rate/Error/Duration) per service
Logs	ELK Stack	Structured JSON logs; MDC traceId propagated; dynamic log level via Actuator
Tracing	Micrometer Tracing + Zipkin	Auto-instrumentation; 10% sample prod, 100% staging
Alerting	Grafana + PagerDuty	SLA breach alerts page on-call SRE; auto-rollback on error rate spike
Deployment	Blue-Green + Canary	Blue-green for major releases; canary at 5%→25%→100% for minor
Chaos Engineering	Chaos Monkey for Spring Boot	Random pod kills in staging; validates Resilience4j circuit breakers work
Load Testing	Gatling	Simulates 50,000 TPS; run weekly in perf environment; results in Grafana

Production Region: AWS ap-south-1 (Mumbai) primary with full active-active deployment in ap-south-2 (Hyderabad) for DR. Route 53 health-check-based failover. RTO < 30 seconds via pre-warmed standby. RPO < 5 seconds via synchronous PostgreSQL streaming replication across AZs. All data stored within India per RBI Payment Data Localisation mandate.

Document Summary

Parameter	Value
Bank Name	Nivesh Bank
Total Microservices	14
Total Domain Entities	103
Total REST API Endpoints	230+
Kafka Topics Defined	21
IFSC Format	NVSH0 + PIN_ZONE(3) + BRANCH_SEQ(3) e.g. NVSH0400001
Account Number Format	11-digit: Type(1) + Region(3) + Sequence(6) + Luhn(1)
Branch Density Rule	Maximum 1 branch per 5km radius — PostGIS enforced
Database Technologies	6 — PostgreSQL, MongoDB, Cassandra, Redis, Elasticsearch, S3
External Integrations	NPCI UPI, RBI NEFT/RTGS, NACH, CIBIL, Experian, UIDAI, NSDL, BBPS, SWIFT, AMFI, Meta WhatsApp, FCM/APNs
Compliance Standards	RBI IT Framework, PCI-DSS L1, PMLA/AML, FATCA/CRS, DPDPA, SEBI Distributor
Target Availability	99.99% (< 52 min downtime/year)
Target Peak Throughput	50,000 TPS
Technology Stack	Java 21, Spring Boot 3.x, Kubernetes, Kafka, PostgreSQL, Redis, Cassandra, MongoDB, Elasticsearch
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This document represents the complete architectural design intent for Nivesh Bank's enterprise microservices platform. All implementation details are subject to technical review, security assessment, and regulatory compliance validation before production deployment.

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